



Software Product Release < *Mutraf* >

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1 Chapter 1: Introduction

In today's fast-paced world, the demand for convenient dining options from food trucks and catering services has surged due to their ability to provide quick and easily accessible meals to busy individuals on the go. However, the process of finding and reserving the perfect food truck or catering service can be challenging, since users tend to waste time during this process given that the available options are few and limited.

Introducing “Mutraf”, a user-friendly mobile application that simplifies the process of reserving food trucks and catering services by connecting users with a wide range of providers in their area. Due to the substantial growth of the food industry and its crucial role in our lives, there is a pressing demand for solutions that can enhance the availability of food to people in a manner that optimizes resources and ensures greater efficiency.

Users can easily find and book food trucks, streamlining the entire experience to support options navigation and informed food decisions.

Importantly, “Mutraf” supports and promotes local food businesses while offering users a convenient way to find vendors that align with their dietary preferences. The application saves users time and effort, ensuring they discover the best food truck to meet their needs [1].

In this document, we will present a significant real-world problem in the food industry and present our solution “Mutraf”. We'll provide a brief overview of “Mutraf” features and user experience. Additionally, we'll discuss its impact on the food industry and the growth of the digital marketplace. We'll present a concise product vision statement, outline the product roadmap, and describe the project objectives at different levels. We'll also define the project scope and introduce the Scrum Team involved, including the Product Owner, Developers, Scrum Master, and Stakeholders. Finally, we'll list the references mentioned for further reading.

In the next section, we will define the problem by exploring the challenges of finding and reserving food trucks and catering services. We will uncover the pain points and highlight the need for a more efficient platform like “Mutraf.”

1.1 The Problem

In this section, we will delineate our problem, which demonstrates the obstacles the users are faced with, and which we aim to solve.

As many people face distractions when planning special events and need help selecting an appropriate food truck, they often waste their time and effort looking at several social media applications, encountering the enormous challenge of identifying food truck accounts and the services they provide. Simultaneously, food truck owners suffer from their inability to find and target customers properly, while also being faced by the difficulty of spreading their services. As a result, people are actively looking for a platform that can simplify the search process and provide complete information for finding and reserving food trucks.

In the next section, we will address how “Mutraf” revolutionizes the way we find and reserve food trucks.

1.2 The Solution

In this section, we will propose the solution to the previously discussed problem. Our solution is a mobile application named “Mutraf”.

“Mutraf” aims to enable smoother experiences for individuals by providing an efficient tool that allows them to easily search for a specific food truck by name, sort them according to multiple criteria,

as well as find them in a Wishlist to locate them easily and confirm the reservation smoothly. Users will be able to edit and delete their reservations as needed. The application also includes a full registration feature, requiring users to create an account. Then, all of the users' information and reservations will be held onto in their accounts, to be safely stored and easily accessible at all times! So, customers can efficiently find their desired trucks without wasting time.

Next, we will present the product vision for "Mutraf." It encompasses the core features and functionality of the mobile application, aiming to provide users with an efficient experience in finding and reserving food trucks.

1.3 The Product

1.3.1 Product Vision

In this section, we will define the product vision of “Mutraf”, which is what we want to achieve *by developing and deploying the app* [2].

To all the people Who want to order food for their occasions inside and outside their houses and want a unique experience for reserving their food, “Mutraf” is a food-truck-reservation Android mobile application That provides a convenient solution for managing food trucks reservations. Unlike to “Jahez” and other food reservation management applications, Our product offers a custom application specifically designed to manage food truck reservations in Riyadh.

In the next section, we will present the Product Roadmap for "Mutraf" that outlines the key milestones and timeline for the development and deployment of the app.

1.3.2 Product Roadmap

In this section, we will show the Roadmap of “Mutraf” app, which demonstrates how the product will be built and delivered over time, as shown in figure 1-1.

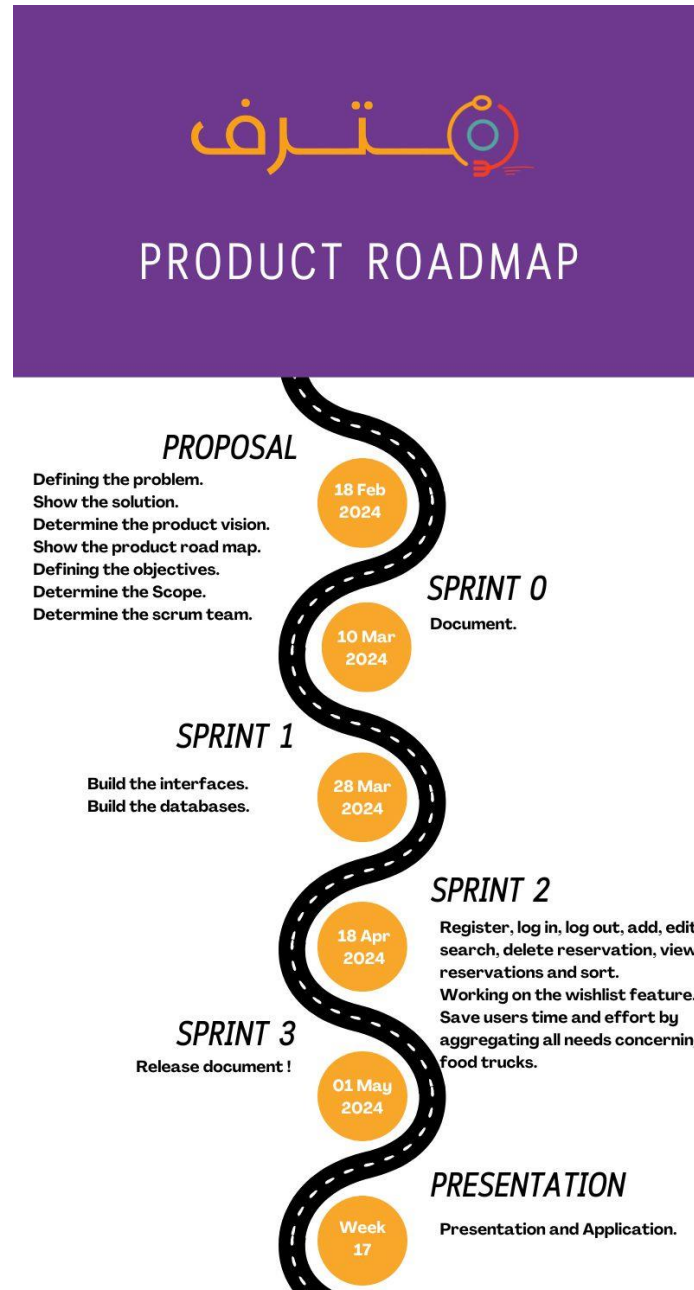


Figure 1-1: Product roadmap

In the next section, we will determine the set of objectives for "Mutraf" app, which defines the requirements that we want to achieve during the process.

1.3.3 Objectives

In this section, we will define the product, project and learning objectives pertaining to the process of “Mutraf” app development. These objectives are the goals and end-state we would like to achieve by the end of this project.

Product Objectives (customer focus-value):

These objectives are related to the product itself, including problems we aim to solve and benefits and features we aim to provide the user with. Main benefits are:

- Present a platform that connects users with many food trucks and catering service providers in their area.
- Save users time and effort by aggregating everything they need concerning food trucks reservations in one app.
- Users can view food trucks details, and search them.
- Users can log-in\out to\from the system, and register as a new user.
- Users can put their desirable\favorite food trucks on a Wishlist.
- Users can add new reservations for any available food truck, edit them and delete them.
- Users can sort the food trucks based on several criteria.

Project Objectives (solution focus-plan):

These objectives are concerned with the stages of developing “Mutraf” app, along with what is going to be achieved by the end of the project. Those are:

- Conduct Interviews and questionnaires.
- Collect and analyze data of food-trucks industry. i.e: Domain analysis.
- Identify user-requirements and apply requirements elicitation.
- Design user-interfaces.
- Design a database associated with “Mutraf” app.
- Develop and program the application and all its functions.
- Test the application.

Learning Objectives (student focus):

These objectives are about the teams’ learning outcomes of this project, including:

- Learn new programming languages, e.g: XML
- Use a new IDE (Flutter Flow)
- Use New tools, e.g: Jira
- Learn and implement Agile app development methodology, and work within scrum team roles.
- Learn and practice data-analysis, user-research and app-testing
- Learn how users’ information is stored within an application through databases.
- Employment of JAVA and other programming languages in mobile application development

By achieving these objectives, our “Mutraf” app will contribute to the growth of the digital marketplace in the food industry while helping our team practice mobile apps development skills.

In the next section, we will discuss the scope of "Mutraf" and delve into the specific functionalities and features it will provide to users.

1.3.4 Scope

In this section, we will define the limitations of our application and what is outside our scope, which will be deemed for future work.

"Mutraf " app is to develop an Android mobile application for making reservations for food truck services in the Riyadh region. The app will be developed using the Java programming language and will support the English language. It will provide users with the ability to view food trucks, add reservations for specific dates and times after checking the availability, as well as delete and edit if needed. Additionally, users will have the ability to search for available food trucks and catering services, sort the options, and menu browsing. An additional function we have is to aid the users with the option to add their favorite food trucks to a Wishlist.

On the other hand, “Mutraf " app will not include features that allow users to rate their experience or contact the food-truck owner. In addition, the functionality of allowing owners to register their food trucks within the system is not included within the app. Any other functions, languages and platforms will not be supported by the end of this project, but will be considered for further work and development in the future.

1.4 The Scrum Team

Table 1-1: Define the scrum team members.

Scrum Team	
Product Owner:	Sarah Alhindi
Developers:	Maryam Altuwaijri Albandry Albakr Sarah Alhindi Wissayif Alkherb
Scrum Master (SM):	I. Ghaida AlFayez
Stakeholders:	I. Ghaida AlFayez

2 Chapter 2: Domain Analysis

In this section, we will introduce the domain in which our customers are expected to use the software in. We will conduct domain analysis to better understand our software and its requirements.

Our domain is the food service industry, it comprises a diverse range of businesses and entities engaged in the processes of preparing, serving, and distributing food and beverages to consumers. This industry includes restaurants, cafes, food trucks, and others. Where their primary goal is to provide meals and dining services to individuals.

The food service industry involves various activities such as food production, food safety and hygiene guarantee, and providing exceptional customer service. Our motivation is to improve existing systems by developing a new and superior solution.

In our application, we will specifically focus on food truck operations and reservations, this sub-part of the food service industry Includes coordination of food vendors, customers, and their interactions within the context of making and managing reservations for food trucks.

In our software engineering process of “*Mutraf*” app, we gathered information from multiple sources, including online resources, e.g. IBISworld.com [1], existing software and its documentation, e.g. HungerStation app [2] to have a better understanding of the software requirements and the rules and regulations governing our domain, along with optimizing the reservation process and enhancing the user experience.

Next, we will provide a list of terms that are relevant to the food service industry and will be used throughout our report and application interfaces to ensure clarity and consistency in communication.

2.1 Terminology

In this section, we will provide a list of some terms commonly used in the food industry domain, as well as in our report and app interfaces.

- **Wish list:** List of items you want, or you like.
In ‘*Mutraf*’ context, food trucks of your interest.
- **Food trucks:** Businesses that specialize in delivering food as well as drinks services at events and celebrations in a truck.
- **Reservations:** The act of scheduling or guaranteeing a time slot or service in advance, typically linked with food services in this context.
- **Food Truck Type:** A classification based on the cuisine or food specialty served by food trucks. Examples include Burgers, Pizza, Ice cream, ...etc.
- **Sorting:** The process of ordering or organizing items according to a specified criterion or attribute. In “*Mutraf*”, it will be based on Price.
- **Event:** An occasion where food trucks may be present to serve customers.
- **Account:** A user's profile containing their information such as name and contact information.
- **SDK:** Software Development Kit, a collection of tools and libraries that developers use to create software applications in a specific ecosystem.
- **Product backlog (PB):** A prioritized list of desired product functionality (product release), that provides a shared understanding of what to build and the order in which to build it.
- **PBI:** Product Backlog Item , a unit of work or a requirement that is added to the product backlog.
- **Use Case:** A descriptive name that reflects a specific functionality in the system.
- **Actor:** A user or an external system interacting with the software.
- **Use Case Diagram:** A diagram used to describe the functional requirements of a software app from the perspective of its users.

2.2 General Domain Knowledge

In this section, we will discuss the important facts and rules in our domain which are normally used and generally known by domain experts.

- Food trucks services: The knowledge of food trucks' rules and ways of operation. This includes how they prepare and serve the food, it also includes understanding types of food trucks, such as Burger truck, Pizza truck, and so on.
- Food Reservation process: The knowledge of reservation process by understanding availability checking, reservation management, conflicts solving, and how users currently reserve food trucks. Any food truck must be associated with its information for user understanding.
- Wishlist Functionality: The knowledge of handling and implementing Wishlist functionality, which enables customers to save and easily access their preferred food trucks for later usage.
- User Experience (UX) Design: Knowledge of the fundamentals and best practices of UX design to provide a simple and interesting user application.
- Food trucks information: Understanding the kinds of data that are usually related to food trucks that would be useful when developing the user interface. Information such as: food truck name, description of the menu, contact details and pictures may be included.
- Data Analysis: knowledge and expertise in data analysis techniques to gather insights from available domain data. This includes knowledge of data visualization and statistical analysis.

2.3 Customers and Users

In this section, we will list and describe users and customers that “*Mutraf*” serves and targets within the food industry, along with some external systems and regulators in the domain. It’s important to note that the customer is the potential buyer/purchaser of software, whereas the user is the potential utilizer/consumer of the software services, while other people in the domain could be other systems, apps and regulators that affect or could be affected by the development and deployment of our app.

Firstly, the customers:

The expected customers of our application are some food service companies in Riyadh, e.g. “ALAMAR FOODS COMPANY”, which is a food company that operates in food service sector and owns many of the restaurants and food service businesses in Riyadh, and Saudi Arabia in general. They would make potential customers buying our software to automate their food-trucks reservation process.

Secondly, the users:

“*Mutraf*” can serve food truck owners and operators, by offering them a platform to exhibit their food trucks and services and be connected to potential customers of theirs. Additionally, the system serves individuals who seek a convenient way to reserve their favorite food trucks and enjoy a delightful dining experience.

We will lastly cover two types of other people who work in the domain, these can be other systems similar to ours, such as the Chefz app, many different types of people work within the Chefz system, including drivers, data analysts, web developers and chefs. Some other people in the domain are regulators such as SFDA (Saudi Food and Drug Authority), they do impose restrictions on food preparation and serving process.

2.4 The Environment

In this section, we will propose the type of “Mutraf” system, followed by the current systems of food-truck reservation, to identify best practices and approaches of reservation. “Mutraf” is a mobile application that will be developed in Flutter Flow environment ver.3.19.6 to work on the Android platform by downloading it from the Google play store. To use the application, users need to have an Android device with a minimum of 19.0 SDK. As for the existing software, like HungerStation and Wssel, they are also mobile applications that work on both IOS and Android platforms. Where in an even simpler system, other food trucks are proposed through social media accounts, e.g. Instagram accounts by the food truck owners themselves.

2.5 Tasks and Procedures

In this section, we will describe what people do with the existing food trucks reservation systems and the procedures they undergo through this process.

It’s important to note that users take different paths to achieve the same goal of reserving a food truck, we will describe the process of reserving through an existing software, e.g. The Chefz and through a social media account, e.g. Instagram.

- People open the “The chefz” app, and search for food trucks from the search box based on the name of the food truck.
- OR
- People open the “Instagram” app, and search for food trucks accounts based on their name (which can be tricky because the username can include punctuation marks, duplicate letters, etc..).
- People check whether the store has a record in Maroof or not, knowing that Maroof is an electronic platform dedicated to introducing electronic stores and providing a statistical measure of the reliability of products and producers.
- People view offers in the food truck page.
- OR
- People view the offers (which can be confusing because they can be viewed in Instagram stories or posts or through bio link).
- People select the preferred offer with any additional options and click “add to basket”.
- OR
- People contact the food truck owner (which can be through Instagram private messages, Instagram comments, calling the owner, or through “WhatsApp” application) to select their preferred offer.
- People select food-truck return time from the order page.
- OR
- People agree on food-truck return time with the owner.
- People proceed to confirm their order and pay via cash or credit card or Apple Pay.
- OR
- People proceed to confirm their order via bank transfer or in cash.
- People receive the food truck and enjoy their dining experience.
- People have their reserved food truck returned and picked up by the owner/The Chefz driver.

As shown, the process can be long and tedious and sometimes even confusing. Thus, we aim to streamline this experience through the development and deployment of ‘*Mutraf*’ app.

2.6 Competing Software

In this section, we will conduct competitive product analysis, this will help us learn about features they offer that we may adopt in our application and identify similarities and differences between our software and the competitors.

We will compare three existing software products that are used in Saudi Arabia (Wssel [3], Mrsool [4], and HungerStation [5]) with our own product “*Mutraf*”. The table includes essential features, strengths, and weaknesses of each competitor. “*Mutraf*” aims to incorporate the strengths of existing solutions while addressing their weaknesses, providing a more comprehensive and user-friendly food truck reservation experience.

Table 2-1: Competing software and their features comparing with *Mutraf*.

Feature	Wssel	Mrsool	HungerStation	Mutraf
View reservations details.	✓	✓	✓	✓
Sort food trucks based on price – Low or High	✓	✗	✗	✓
Delete a reservation	✗	✓	✗	✓
Support Arabic language	✓	✓	✓	✗
Wish list	✗	✗	✗	✓

2.7 Similarities Across Domains and Organizations

In this section, we will explore the distinctiveness of the food service industry and its differentiation from other domains. Additionally, we will highlight the similarities that exist between the food service industry and related sectors.

The food service industry sets itself apart from other domains through its distinct characteristics and objectives. It revolves around the preparation, serving, and distribution of food and beverages to cater to individuals' culinary needs and provide dining experiences. This distinguishes it from sectors like manufacturing or retail, which have different primary functions and goals.

In the context of the food service industry, the domain-specific focus of the food truck reservation management app “*Mutraf*” further sets it apart. This specialized application specifically targets the operations and reservations of food trucks. By coordinating food vendors, managing reservations, and facilitating interactions between customers and food service providers, and providing extra functionalities like Wishlist option, “*Mutraf*” caters to the unique requirements of this sub-part of the industry.

Despite these distinctions, the food service industry also shares similarities with other domains. One commonality is the emphasis on maintaining high standards of food quality and safety. This applies to all sectors within the food service industry, including food truck operations, and is also a crucial aspect in other industries that handle food products. Additionally, customer satisfaction and delivering excellent service are aspects that the food service industry shares with various other domains, such as hospitality and retail, as both prioritize providing a positive customer experience and addressing customer needs.

In summary, the food service industry, with its specific objectives and characteristics, differentiates itself from other domains. The food truck reservation management app "*Mutraf*" aligns with this domain, catering to the unique requirements of food truck operations. However, commonalities such as a focus on food safety and customer satisfaction also exist between the food service industry and other related domains.

3 Chapter 3: Requirements Engineering

In this section, we will discuss key phases of the requirements engineering process. Firstly, we documented and elicited requirements using techniques like user interviews and questionnaires. Also, the characteristics of the system's users were explored, and a Use Case Diagram was created to visualize the app's functionality. In addition, our report included a Product Backlog composed of backlog items. Finally, User stories were used to concisely articulate desired functionalities.

As part of our requirements engineering process for the "*Mutraf*" app, we began by documenting the requirements. To ensure a comprehensive understanding of user's needs, various methods can be employed to explore requirements, such as observation, workshops, focus groups, use cases and prototypes. However, in our case, we utilized two specific approaches for requirements elicitation: user interviews and questionnaires. We chose these methods because they offered the most suitable fit for our needs, and we did not exhibit any conflicting requirements. For the user interviews (See Appendix A: Interview), we conducted interviews with two stakeholders as a valuable source of information. We formulated five targeted questions that addressed their perspectives and expectations regarding the app's functionality, usability, difficulties, and features. Through these interviews, we gained insights into their specific requirements by collecting their feedback and responses in the interview's transcriptions. These transcriptions encompass all interview details, such as names, specific locations, time, date, objectives, agenda, as well as notes and topics that were not covered during the interview. This comprehensive collection of information allows us to shape the application in the most optimal manner. Additionally, we utilized questionnaires (See Appendix B: Questionnaires) as another method of requirement's elicitation. We distributed a set of seven carefully crafted questions to a wider group of stakeholders, collecting a total of 22 responses. The questionnaire covered various aspects, such as desired features, and potential challenges. By analyzing the collected data and converting it to pie charts, we gained a deeper understanding of the stakeholders' collective needs and preferences, allowing us to align the development process with their expectations.

The interviews and questionnaires conducted provided valuable insights and findings regarding the preferences and needs of individuals when it comes to food trucks and related services. Based on the interviews, it was found that guests enjoy buffet-style setups for special events, allowing them to choose from a variety of dishes and customize their plates. However, another preference was having food stations where guests could interact directly with food truck Chefs for freshly prepared meals.

When it comes to difficulties in finding a good food truck, the interviewees mentioned challenges such as aligning with desired cuisine and theme, as well as accommodating dietary restrictions or special requests from guests. Recommendations from friends or event planners, as well as online research and vendor contact, were common approaches for selecting food trucks. In terms of cancellation policies, clear communication and fair refund policies were expected by both interviewees. A straightforward and transparent cancellation procedure was deemed important. Additionally, there was a strong interest in using an application that brings together all food trucks, with features such as pre-reservation, sorting options based on prices, and a Wishlist for food trucks.

As for the questionnaire, the majority of respondents (59.1%) indicated that finding a good food truck for their needs, such as the type of food and budget, is not easy nowadays (as shown in Figure 6). This highlights the challenges faced when seeking a suitable food truck for events. Furthermore, a significant percentage (90.9%) of respondents expressed a preference for food delivery services,

indicating a strong demand for convenient and hassle-free options (as shown in Figure 7). Similarly, 90.9% of respondents showed interest in using an application that offers all food trucks in one place (as shown in Figure 8), emphasizing the desire for a centralized platform to access a wide range of food truck options. The findings also revealed that a considerable percentage (86.4%) of respondents would appreciate the option to pre-reserve food trucks (as shown in Figure 9), allowing them to save time and avoid long lines. This reflects the importance of efficient and streamlined processes in the food truck experience. Additionally, the majority of respondents (81.8%) expressed enjoyment in trying different cuisines from a variety of food trucks in Riyadh (as shown in Figure 10). This highlights the appeal of diverse culinary experiences offered by food trucks. Regarding the organization and search process, 81.8% of respondents indicated a desire for the option to sort food trucks based on prices (as shown in Figure 11), this feature would assist them in finding food trucks that align with their budgetary considerations. Moreover, 77.3% of respondents expressed interest in a "Wishlist" feature for food trucks (as shown in Figure 12), further emphasizing the need for enhanced organization and time-saving tools. Overall, the results highlight the importance of variety, convenience, and customization in the food truck reservation experience. The findings provide valuable insights for the development of an application that caters to the needs and preferences of users seeking food truck services in Riyadh. (See Appendix A: Interview) (See Appendix B: Questionnaires)

In summary, for the requirements engineering process. In the beginning, we focused on documentation and requirement elicitation, employing techniques such as user interviews and questionnaires to capture and document specific app requirements. Also, the System's Users section described the general characteristics of our users, including their educational level, experience, and technical expertise. This understanding allowed us to tailor the app to meet their specific needs. We also created a Use Case Diagram, providing a visual representation of the app's functionality from the user's perspective. This diagram illustrated the various interactions and use cases supported by the system. In addition, our report included a Product Backlog, composed of backlog items or PBIs. These PBIs represented features and functionalities that would deliver tangible value to the app's users or customers. Finally, user stories were used to concisely articulate these desired functionalities.

3.1 System Users

This section describes general characteristics of our users including educational level, experience, and technical expertise.

For our application, we have only one type of users who make reservation; they can access the application and utilize its features to book their desired food trucks.

- Users should be able to read/write and be familiar with English words and letters.
- There is no specific educational level for our users, they can have diverse educational levels as long as they are adults and responsible.
- Users should have minimal technical knowledge to navigate and utilize the features effectively.
- Users are expected to have basic familiarity with similar applications in general.
- Users should be interested in exploring different food options and enthusiastic about experiencing food trucks. With a need to make reservations.
- Users are expected to be +18 years old, i.e: Adults.
- Users are of both genders, i.e: Female and Male

3.2 Use Case Diagram

In this section, we will provide an overview of the system's functionality and interactions with its actors (user or external system) and use cases (functionality). It's important to note that our users can do all the functionalities directly with no need to have an extends or include relationships.

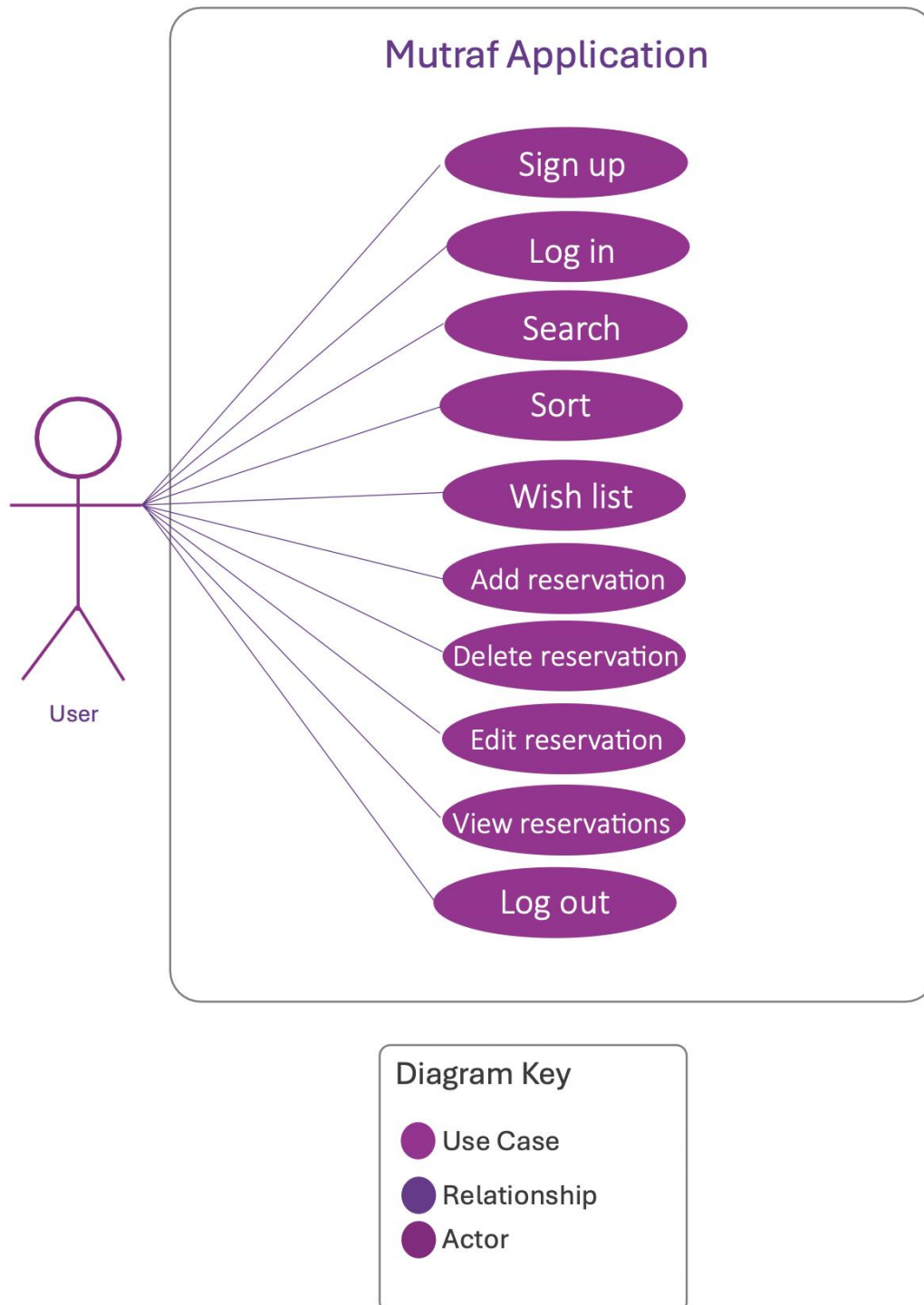


Table 3-1: Use case diagram of Mutraf Application.

3.3 Product Backlog

In this section, we will provide the product backlog, The product backlog is composed of backlog items (PBIs), and usually takes the form of a table with four columns, the PBI, its size, type and acceptance criteria, ordered according to their importance “from high to low priority”. The building blocks of the product backlog are user stories. User stories are a common format for representing items of business value” [6]. This product backlog will also be represented in Jira.

The product backlog is composed of backlog items (PBIs). “PBIs are features, items of functionality that will have tangible value to the user or customer. These are often written as user stories.

Table 3-2: Definition of Ready for Mutraf user stories

Example Definition of Ready	
✓	Business value is clearly articulated
✓	Details are sufficiently understood
✓	Dependencies are identified; no blocking dependencies exist
✓	Team is appropriately staffed relative to the PBI
✓	Estimated and small enough to be completed during sprint
✓	Acceptance criteria are clear and testable
✓	Performance criteria, if any, are defined and testable
✓	Team understands how to demo the completed PBI

Table 3-3: Product backlog of Mutraf.

PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Acceptance Criteria The conditions of satisfaction that must be met for that item to be accepted.
As a user, I want to be able to register and sign up so that I can use the application and its features.	3	Feature	- As a nonregistered user who wants to sign-up, if I go to signup page and enter my username, password that is more than 8 numbers, and phone number and click on sign up button, then I should be able to access the application. - As a nonregistered user if I go to the sign-up page and enter a phone number or username that already exists and click on sign up button, then I should receive an error message. - As a non-registered user who wants to sign up, if I leave any required fields blank (e.g., username, password, phoneNO), the system should display an error message since all fields are mandatory and prompt me to fill in the missing information.
As a user, I want to log in with my username and password so that the system can authenticate me, and I can access my account and my information.	2	Feature	- If I go to the log-in page and enter my username and password and click on Log in, then the data associated to my user should be accessible. - If I go to the log in page and enter incorrect username or password and click on Log in, then log in fails with an error message that shows the username or password was wrong without specify which is wrong for security. - If I go to the log in page and enter my username and password and click on Log in, then my user login session is loaded in less than eight seconds.

As a user, I want to log out from my account so that I ensure the security of my account and personal information.	2	Feature	- As a logged-in user, if I click on the "Logout" button within the application, then I should be successfully logged out from my account. - If I logged out and try to access any protected pages or features within the application, then I should be redirected to the login page, indicating that I no longer have access until I log back in.
As a user, I want to search for specific food trucks by its name so that I can easily find and access my preferred food truck.	2	Feature	- As a user, when I access the food truck search section in the app, I should be presented with a search bar or input field. - If I submit the search query, then the system should process the request and display the search results, focusing on food trucks that match the entered name. - If I submit the search, then the search results should prioritize exact matches or closely related matches to the entered name, ensuring that the preferred food truck is prominently displayed. - If there are no food trucks matching the search criteria, then the system should display a message indicating that no results were found.
As a user, I want to sort food trucks based on its price so that I can find the most suitable option for my budget easily.	2	Feature	- As a user, if I access the food truck listing in the app, then I should be presented with price sorting option with two options (from lowest- from highest) - If I select a specific sorting option, then the system should process the request and rearrange the food trucks accordingly. - As a user, if I finish exploring the result of the sorting, then I should have the ability to reset or clear the sorting criteria to return to the default listing order.

As a user, I want to create a Wishlist , so that I can conveniently keep track of my preferred options and quickly access them when needed.	1	Feature	- As a user, if I access the food truck listing in the app, then I should have the option to add food trucks to my Wishlist. For each food truck, there should be a button or icon that allows me to add it to my Wishlist. - If I click on the Wishlist button/icon for a specific food truck, then the system should add it to my Wishlist and provide visual feedback (e.g., change in color or appearance) to indicate that it has been successfully added. - If I receive the visual feedback, then the system should provide a dedicated section or page where I can access and manage my Wishlist that I have added, including relevant trucks information. - As a user, if I start the application, then I should be able to view and navigate to my Wishlist at any time and from any page within the application. - If I click at the heart button again, then the system should remove food trucks from my Wishlist with providing a visual feedback.
As a user, I want to make a reservation for a food truck or catering service of my choice so that I can secure dining arrangements and ensure availability	3	Feature	- As a user, if I access reservation section, then I should be presented with list of available food trucks or catering services. - If I have found a food truck or catering service, then I should be able to view detailed information about it, including menu options, pricing, and available dates/times for reservation. - As a user, if I select a specific date and time for my reservation, then the system should check the availability of the chosen food truck or catering service and display confirmation if it is available. - If the chosen food truck is not available for the selected date and time, then the system should provide a message indicating that the food truck is not available. - If the reservation is successfully made, then the system should provide me a confirmation message.

As a user, I want to edit my existing reservation details (e.g., date, time) so that I can reflect necessary changes of my dining plans.	2	Feature	- As a user, if I go to reservation page, then the reservations should display with a button to edit the reservation details. - If I select the edit option for a reservation, then the system should display the current details of the reservation, including the date, time, food truck name, and any additional information. - If I go to the reservation page, then the system should allow me to modify the existing reservation details, such as the date, time. - If the modified reservation details are valid and available, then the system should update the reservation with the new information - If the reservation is modified, then the system should provide a confirmation message indicating that the changes have been saved. - If the modified reservation details are not valid or conflict with existing reservations, then the system should display an appropriate error message and prompt me to select alternative options or adjust the changes.
As a user, I want to delete my reservation if my plans change, so that I can free up the reserved slot and allow other users to make use of it.	2	Feature	- As a user, if I access my account, then I should be able to view a list of my existing reservations - If i go to the reservations in my account, them the system should displaye all reservations with a button to delete them. - If I select the delete option for a reservation, then the system should prompt for confirmation to ensure I intend to cancel the reservation. - If I confirm the deletion, then the system should remove the reservation from my account and free up the reserved slot for other users. - If i click on delete button, then the system should display a confirmation message indicating that the reservation has been successfully canceled.

As a user, I want to view detailed information about food trucks, such as their cuisine and available hours, so that I can make informed decisions and choose the best option for my dining experience.	1	Feature	- As a user, if I access the food truck listing from the app, then I should be able to click on a specific food truck to view its detailed information. - If I go to the detailed information [age, then the system should display relevant details about the food truck, including its name, cuisine type, contact information, and any additional information that helps users identify and understand the food truck.
As a user, I want to view my reservations within the application, including details such as the date, time, food truck name so that I can stay organized and keep track of my upcoming dining arrangements.	1	Feature	- As a user, if I access my account, then I should be able to see a list of my reservations. - If I go to the reservations page, then the system should display essential information for each reservation, including the date, time, and food truck name. - If I select a reservation from the list, then the system should provide a detailed view that includes all pertinent information about the reservation. - If there are any updates or changes to a reservation, such as a modified date or time, then the system should reflect the updated information in the reservation view. - As a user, if I go to reservations page, then I should be able to navigate between different reservations and switch between the list view and detailed view effortlessly with a clear and organized manner.
As a user, I want the system to be available 95% of the time when I try to access it, so that I don't get forced to search for other applications to use.	5	Feature	As a user, if I opened the application 10 times a day, then 9 of them should be available to use.
As a user, I want the search results for available food trucks to be displayed within 5 seconds after submitting desired option, ensuring a fast and responsive experience.	3	Feature	If I submit a search query for food trucks, then the system should process the request and display the search results within 5 seconds.

3.4 Definition of Done

Table 3-4: Definition of Done for Mutraf user stories.

Example Definition of Done	
✓	Design Interfaces
✓	Interfaces reviewed and approved to meet UX/UI guidelines
✓	Code reviewed and no bugs found
✓	Documentation has been created and updated as necessary
✓	User acceptance testing has been conducted
✓	Integration testing has been conducted
✓	User Story acceptance testing has been conducted

4 Chapter 4: System Design

The System Design section consists of five components, each of which contributes to the overall design of our system.

First, we describe the selected System Architecture and provide a diagram of it. Second, we provide a thorough class diagram. Third, we illustrate Data Design using an ER Diagram, Schema, and Data Dictionary. Fourth, we provide Component Design with pseudocode and effort estimates for adding, editing, and deleting reservation functions. Finally, we demonstrate Interface Design using a navigation diagram, screen images, and five UX recommendations that were implemented.

These components include our 'Mutraf' architecture, class structure, data management, component functionality, and user interface design, all of which combine to provide a comprehensive and user-friendly solution.

4.1 System Architecture

The system architecture of an application affects its overall performance, scalability, and maintainability. In this section, we will talk about the system architecture we chose for our application "Mutraf".

The system architecture is the basis of the whole program, outlining how various components interact and communicate with one another in order to deliver the desired functionality and user experience. Following extensive investigation and evaluation of several design patterns, we chose the Client-Server architecture for our application (Figure 4.1). The Client-Server Architecture separates the system into two main parts: client-side and server-side.

The client-side component provides a user-friendly interface for registration, login, seeing vehicle data, and making reservations. It provides a user-friendly experience by successfully displaying essential information, effectively capturing user input, and presenting data in an easy-to-use way.

In contrast, the server-side component is critical to the design since it serves as a central database for storing important data about users and trucks. It also includes the application's business logic. The server side is responsible for responding to user requests, performing necessary computations, retrieving and updating data from the database, ensuring that users have access to correct and up-to-date data and tasks, and maintaining the application's data. Using the programming language Dart.

We chose Client-Server Architecture for a variety of reasons. First, it distinguishes between the display layer (client-side) and the business logic and data layer (server-side). This separation improves the application's modularity and maintainability, allowing for easy updates and improvements to certain components without impacting others. Scalability is one of the advantages of the Client-Server Architecture. It lets several clients to connect to the server at the same time, making it easier to handle a big user data and a high volume of requests. Additionally, the client-server Architecture improves security. By concentrating data and business logic on a server, security measures like authentication and access control become simpler, protecting sensitive information and allowing only authorized users to access and change it.

We investigated multiple architectural possibilities for the "Mutraf" application but eventually disregarded them for a variety of reasons. The MVC (Model-View-Controller) Architecture, which is widely used in web-based applications, was not chosen because it is better suited for complex user interfaces and many views, which would add additional complexity and overhead to our modest system. The Layered Architecture, which is commonly used in systems with various development teams, was not chosen because it is better suited to systems that are ready but require more development, and its multiple levels of abstraction may have an influence on performance. However, we did not chose the Repository Architecture that is widely used that is manage data access and storage in complex systems, and our system does not necessitate it which will make unnecessary overhead. Lastly, Pipe and Filter Architecture is used in data processing systems, without any interaction with the user; that is why it is not suitable for our system.

In conclusion, following thorough consideration of the particular requirements of the "Mutraf" application, we determined that the Client-Server Architecture was the best option. It offers the essential separation of concerns, scalability, and security to ensure the application's effective deployment.

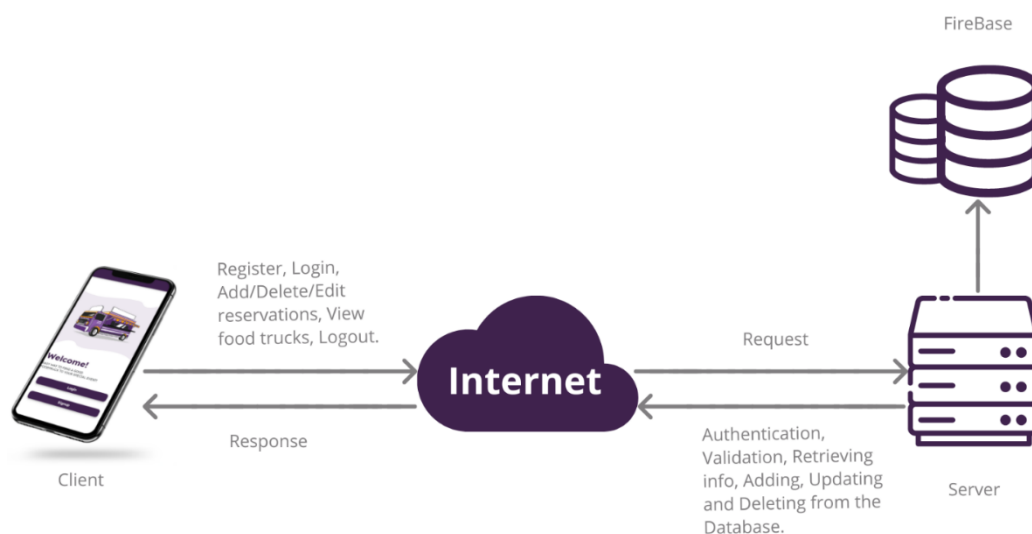


Figure 4-1: System Architecture of Mutraf.

4.3 Data Design

In this segment, we will explore the design of a data system. We will highlight three key elements: the ER Diagram, the Schema, and the Data Dictionary. The Entity Relationship Diagram illustrates the entities, attributes, and relationships within our database. The schema outlines the structure of our database, including tables and their relationships. The Data Dictionary organizes and provides descriptions for entities, attributes, and relationships. By employing these components, we ensure that our system's data is appropriately organized. ER Diagram.

- ER

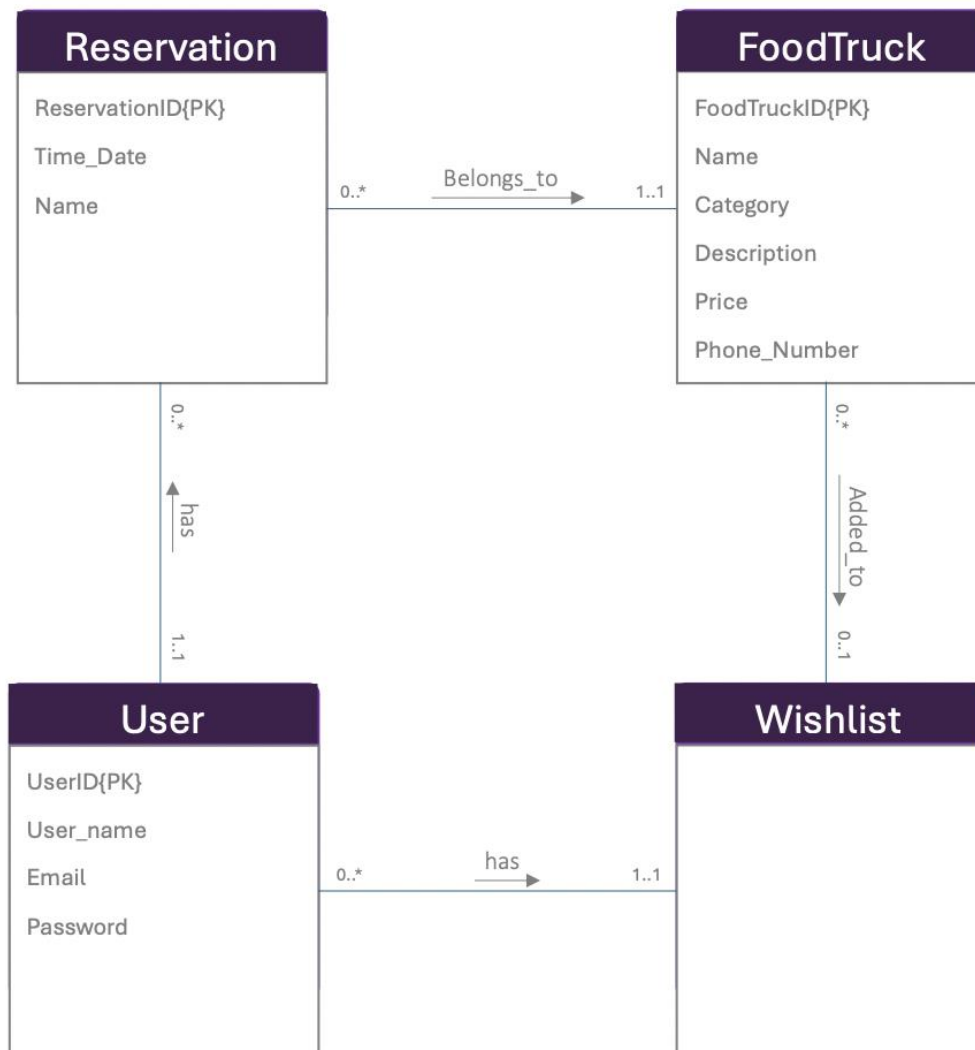


Figure 4-3: ER Diagram of Mutraf Application.

- Schema

User (UserID, Email, Password, User_name)

Primary Key: UserID

FoodTruck (FoodTruckID, Name, Category, Description, Price, Phone_Number)

Primary Key : FoodTruckID

Reservation (ReservationID, FoodTruckID, UserID, Time_Date, Name)

Primary Key: ReservationID

Foreign Key: FoodTruckID References FoodTruck (FoodTruckID)

Foreign Key: UserID References User (UserID)

Wishlist (FoodTruckID)

Foreign Key: FoodTruckID References FoodTruck (FoodTruckID)

- Data Dictionary:
- Data Dictionary showing description of all entities.

Table 4-1: Data Dictionary showing description of all entities.

Entity Name	Description	Aliases	Occurrence
User	A User refers to any person who makes use of our food truck reservation management application, which are the individuals who are interested in reserving a food truck from another party.	-	A user can have none or several reservations. A user has exactly one wishlist.
FoodTruck	A food truck, within the context of our application, is a truck that offers food for events that is available for reservations, in spite of its type or cuisine.	-	A food truck may belong to multiple reservations. A food truck can be added to exactly one Wishlist for a user.
Reservation	A reservation in our application represents a food truck that has been booked by a user	-	A reservation is belonging to exactly one food truck. A reservation belongs to a single user.
Wishlist	A wishlist is a collection of food trucks that users have chosen and saved for future reference or potential reservations.	-	A wishlist is uniquely associated with one user. A wishlist can contain several food trucks, or potentially none.

- Data Dictionary showing description of all relationships:

Table 4-2:Data Dictionary of all Mutaf's relationship.

Entity Name	Multiplicity	Relationship	Entity Name	Multiplicity
User	1..1 0..*	Has Has	Reservation Wishlist	0..* 1..1
FoodTruck	1..1 0..*	Belongs_to Added to	Reservation Wishlist	0..* 0..1

- Data Dictionary showing description of all attributes:

Table 4-3:Data Dictionary showing description of all attribute.

Entity Name	Attribute	Description	Data Type	Length	Nuls	Mult i-Valued	Default Value	Range	P K
User	UserID	User's Id	String	255	N	N	N		Y
	User_name	User's name	String	30	N	N	N		
	Email	User's email	String	255	N	N	N		
	Password	User's password	String	100	N	N	N	6-255	
FoodTruck	FoodTruck ID	Food truck's Id	String	255	N	N	N		Y
	Category	Food truck's type	String	255	N	N	N		
	Description	Food truck's details	String	255	N	N	N		
	Price	Food truck's price for a reservation	double		N	N	N		
	Phone_Number	Food truck's phone number for contact	String	10	N	N	N		

	Name	Food truck's name	String	30	N	N	N		
Reserva tion	Reservatio nID	Reservati on's Id	String	255	N	N	N		Y
	Time_Date	Reservati on's date and time	timeD ate		N	N	12:1 2		
	Name	Reservati on's name	String	30	N	N	N		

4.4 Component Design

In this section, we will look at our system's component design, focusing on the functionality for adding, editing, and deleting reservations. We will give design specifications and pseudocode for each user story, defining the logic and steps necessary for executing them. In addition, we will estimate the time and effort needed for each user story. We aim to give a complete knowledge of the implementation process for these reservation services by giving thorough design information, pseudocode, and time estimates.

Table 4-4 :Component Design.

Sprint number	PBI (user story)	User story tasks with effort estimates in hours				Sum of effort
2	As a user, I want to make a reservation for a food truck or catering service of my choice so that I can secure dining arrangements and ensure availability	Hours: 4	Hours: 3	Hours: 2	Hours: 2	11
		Implement a "Reservation" class to hold reservation information and a new table in the database to store the reservation information.	Design "Food truck" interface including button for date and time choosing, and reserve button.	Create a button in the "Food truck" page to select date and time from pop-up window for adding the reservation.	Create confirmation message, in case of success, adding it to reservation table then display message, and display the reservation in "My reservations" interface.	

2	As a user, I want to edit my existing reservation details (e.g., date, time) so that I can reflect necessary changes of my dining plans.	Hours: 3	Hours: 1	Hours: 3	Hours: 2	9
		Implement a new function to edit the reservation's date and time from the database.	Implement a "Edit" button on the "My Reservation" page adjacent to each reservation.	Design "Edit reservation" interface including button for date and time choosing, and update button.	Create confirmation message, in case of success, edit date and time in the reservation, and display the change in "My reservations" interface.	
2	As a user, I want to delete my reservation if my plans change, so that I can free up the reserved slot and allow other users to make use of it.	Hours: 3	Hours: 3	Hours: 1		7
		Implement a new function to delete the reservation from the database.	Implement a "Delete" button on the "My Reservation" page adjacent to each reservation. When the button is clicked, a confirmation message appears, and linked the button to the corresponding function.	Checking deletion of the Food truck from the database		

- **Add Reservation:**

Classification: Function

Definition:

The user should be able to add a new reservation for food truck by selecting all necessary information.

Construction:

Inputs: Date, Time.

Precondition: User must be signed in.

Postcondition: The reservation is added in the database and displayed in "My reservations".

Pseudocode:

BEGIN

IF user clicks on a food truck in home page:

DISPLAY food truck page with food truck details (name, price, phone number, category, description)

DISPLAY "Choose" button

DISPLAY "Reserve" button

IF user clicks on "Choose" button:

DISPLAY pop-up window for selecting date and time

ENDIF

IF user clicks on "Reserve" button:

 Add reservation information to the database

DISPLAY message to confirm the addition

DISPLAY message to inform the user the reservation is done

DISPLAY the reservation in "My reservation" page

ENDIF

ENDIF

END

- **Edit Reservation:**

Classification: Function

Definition:

The user should be able to edit a reservation for food truck by selecting all necessary information.

Construction:

Inputs: Date, Time.

Precondition: User must be signed in and the reservation must already be added.

Postcondition: The reservation is updated in the database and displayed the change in "My reservations".

Pseudocode:

BEGIN

IF user clicks on the "Edit" button in "My Reservations" page:

DISPLAY the "Edit Reservation" page

DISPLAY "Choose" button

DISPLAY "Update" button

IF user clicks on "Choose" button:

DISPLAY pop-up window for selecting date and time

ENDIF

IF user clicks on "Update" button:

Update the reservation information in the database

DISPLAY message to confirm the edition

DISPLAY message to inform the user the reservation is updated

DISPLAY the updated reservation in "My Reservations" page

ENDIF

ENDIF

END

- **Delete Reservation:**

Classification: Function

Definition:

The user should be able to delete their food truck reservations.

Construction:

Precondition: The user must be signed in and the reservation must already be added.

Postcondition: The reservation is deleted from the database and removed from "My reservations".

Pseudocode:

BEGIN

IF the user clicks the "Delete" button in "My Reservations" page:

DISPLAY a message to the user asking him to confirm the deletion.

IF the user confirms the deletion:

Delete the reservation from the database

Delete the vehicle from the "My reservations" page

DISPLAY message to inform the user the delete is done

ELSE

Do nothing

ENDIF

ENDIF

END

4.5 Interface Design

This section will focus on our system's Interface Design. We will use a navigation diagram to describe the application structure, emphasizing on the flow between screens and presenting the general architecture of our interface. Furthermore, we will give screenshots with descriptions of the important fields for a better understanding of the interface's functionality. Lastly, we will demonstrate our commitment to a user-centered approach by giving five UX standards that were incorporated into the design process of our app.

- **User Navigation Hierarchy:**

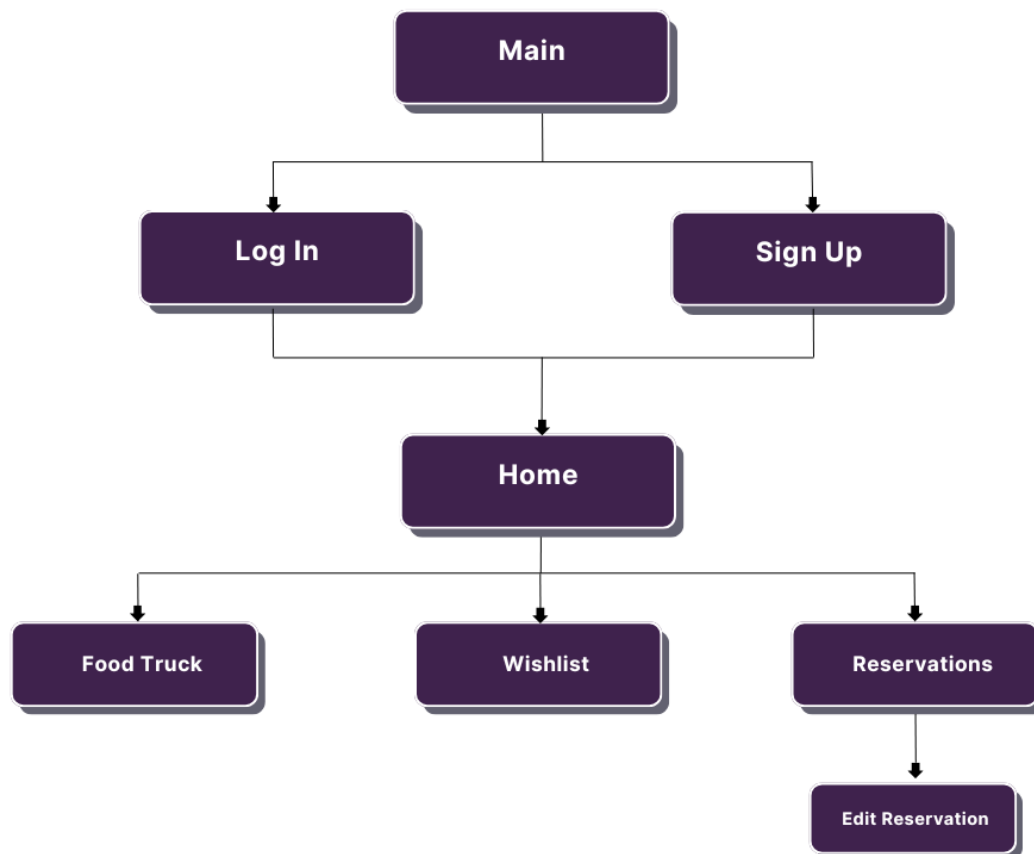


Figure 4-4: User Navigation Hierarchy

- **UX Guidelines implemented for each interface:**

- 1. Consistency design.**

Guarantees clients have a consistent and comfortable experience across the app. It involves using standardized visual components—like color palettes, fonts, and layouts—across different screens.

- 2. Clear Navigation.**

Using intuitive and easily recognizable navigation elements, to guide users to different sections or features of the app. We also provide clear labels for navigation options like icons in the bottom navigation bar.

- 3. Feedback and Confirmation.**

Offer immediate feedback to acknowledge user actions and provide confirmation of completed tasks, such as when the user wants to delete a truck , he will be asked for confirmation then will be presented with a “deleted successfully” message if he proceeds with the process.

- 4. Flexibility and Efficiency of use.**

Provide streamlined workflows for quick and efficient task completion. Such as the seamless and quick flow of switching between the pages in the app.

- 5. Error Handling.**

Prevent errors where possible by validating user inputs and providing clear instructions. When errors occur, informative error messages should be displayed, explaining the issue and suggesting possible solutions, such as when the user enters an invalid email format in the registration process.

- User interfaces:

1. Main page:

Table 4-5: Main page description.

#	Description
1	Log in button that direct the user to log in page.
2	Sign up button that direct the user to sign up page.

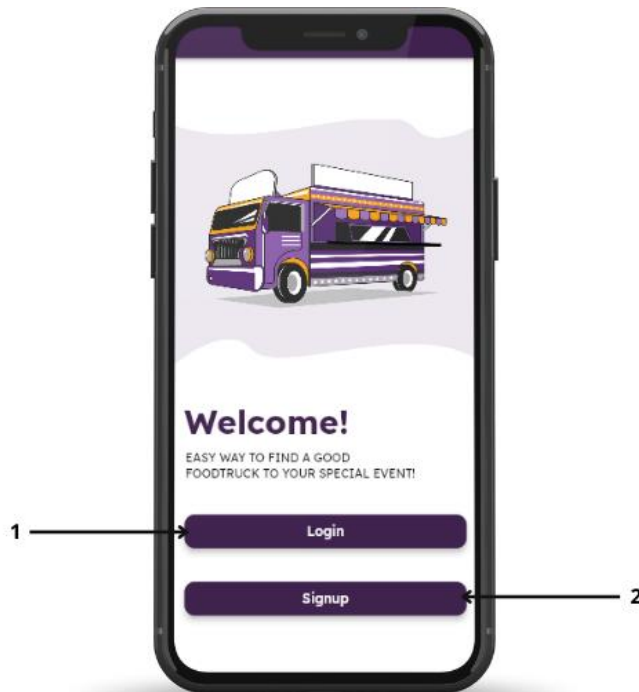


Figure 4-5: Main page

2. Sign up page (if user click on #2):

Table 4-6: Sign up page description.

#	Description
3	User's email input field that is mandatory to register as new user and access user's page.
4	User's name input field that is mandatory to register as new user and access user's page.
5	User's password input field that is mandatory to register as new user and access user's page, and it must be at least 6 digits.
6	User's confirmation password input field that is mandatory to register as new user and access user's page, and it must be like the password on the past field.
7	Log in button that transfer user to log in page if already have an account.
8	Create account button that transfer the new user to home page when all information entered.

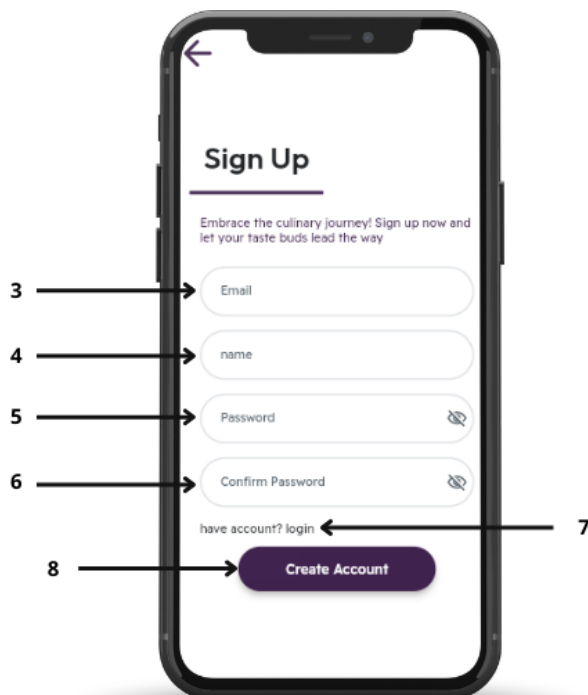


Figure 4-6: Sign up page.

3. Log in page (if user click on #1):



Figure 4-7: Log in page

Table 4-7: Log in page.

#	Description
9	User's email input field that is mandatory to access user's page.
10	User's password input field that is mandatory to access user's page.
11	Signup button that transfers the new user to sign up page.
12	Log in button that transfer the user to the home page when all information entered.

4. Home page (if user signup/login successfully):

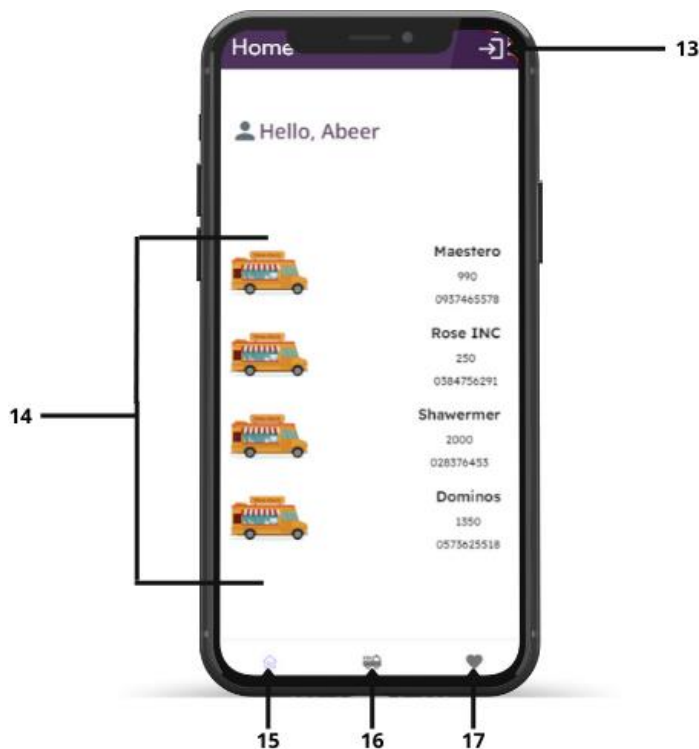


Figure 4-8: Home page.

Table 4-8: Home page description

#	Description
13	Log out button to direct the user to the main page.
14	Displayed the food trucks with its name, price and phone number, and the user has the ability to select the food truck they want to reserve.
15	Home button to direct the user to the home page.
16	Truck button to direct the user to 'My Reservations' and the user will be able to view the food trucks that he has been reserved.
17	Wishlist button direct the user to Wishlist page.

5. Food Truck View\Reserve page (if user click on #14):

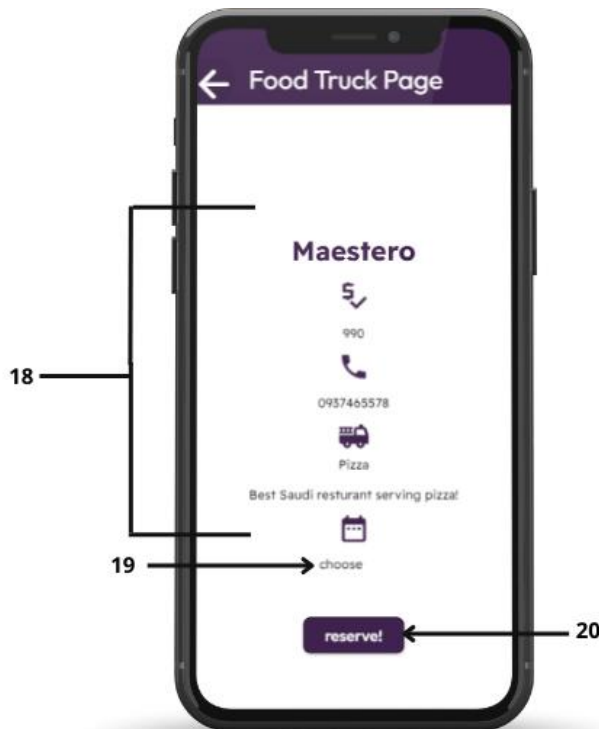


Table 4-9: Food Truck page description

#	Description
18	The user has the ability to view the details of a food truck before reserving it.
19	The user has the ability to click on 'choose' button that pop-ups a window for user to select date and time for his reservation.
20	The user has the ability to reserve a food truck by clicking on 'reserve!' button, then shows confirmation messages to make sure the user wants to reserve or not.

Figure 4-9: Food Truck page.

6. Food Truck page 'Select Date pop-up window' (if user click on #19):



Table 4-10: Food Truck page description.

#	Description
21	The user has the ability to view and select date for his reservation.
22	Clicking on Cancel will close the pop-up window without choosing date and time.
23	Clicking on Ok will drive the user to Select time pop-up window.

Figure 4-10: Food Truck page (Select Time pop-up window).

7. Food Truck page 'Select Time pop-up window' (if user click on #23):

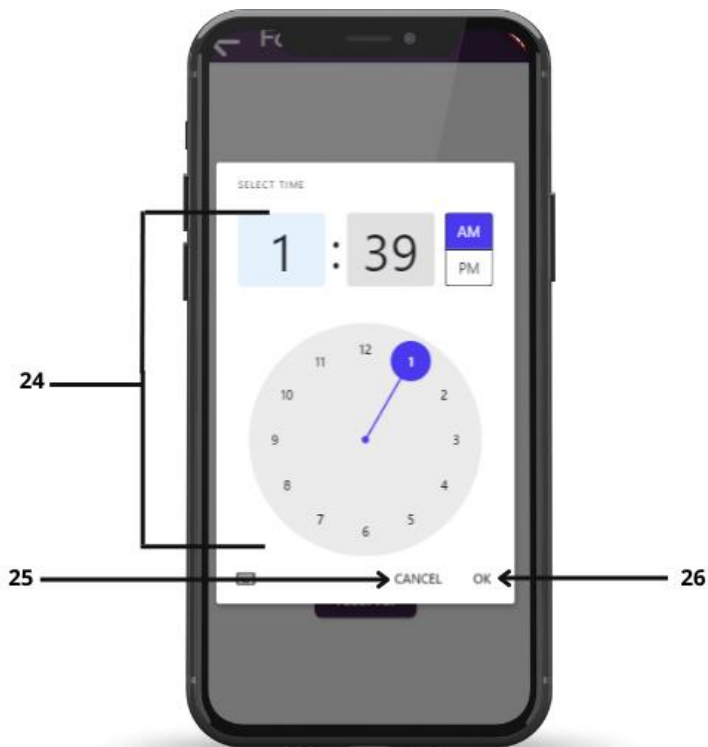


Table 4-11: Select Time pop-up window description.

#	Description
24	The user has the ability to view and select time for his reservation.
25	Clicking on Cancel will close the pop-up window without choosing time.
26	Clicking on Ok will close the pop-up window with choosing the date and time.

Figure 4-11: Food Truck page (Select Time pop-up window).

8. Food Truck page 'Reservation Confirmation Message' (if user click on #20):

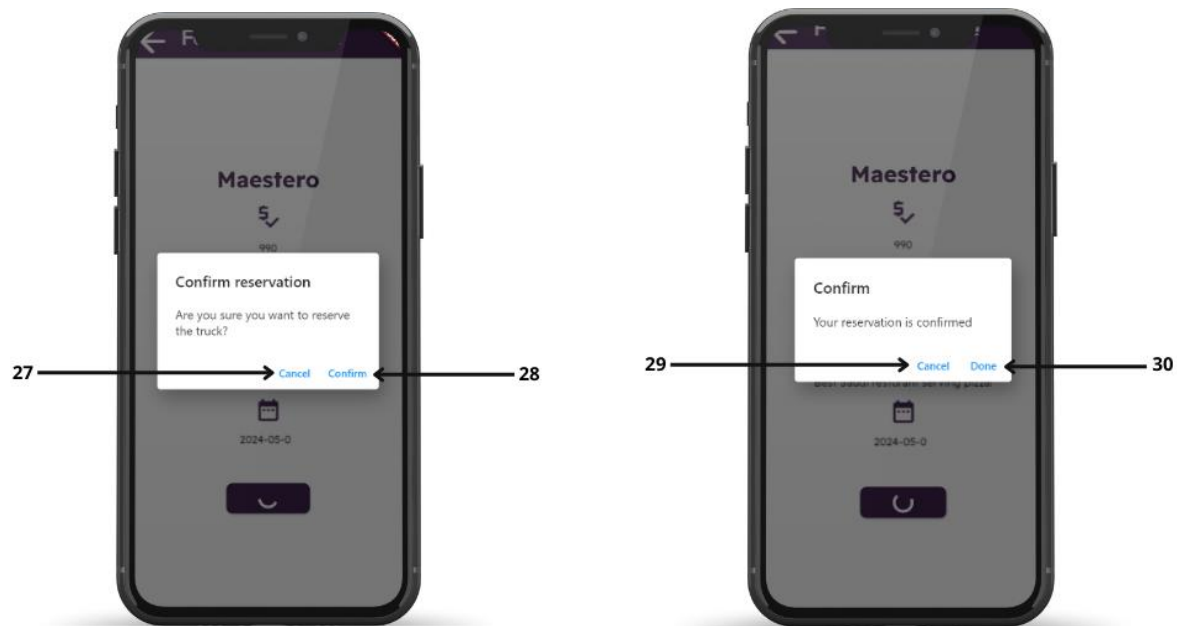


Figure 4-12: Reservation confirmation message.

Table 4-12: Reservation confirmation message description.

#	Description
27	Clicking on Cancel will close the pop-up window without reserving.
28	Clicking on Confirm will confirm the reservation for the user and add it to 'My Reservation' page and opening another confirm pop-up window.
29	Clicking on Cancel will close the pop-up window.
30	Clicking on Done will close the pop-up window with reserving.

9. My Reservations Edit/Delete/Search page (if user click on #16):

Table 4-13: Edit/Delete/Search description.

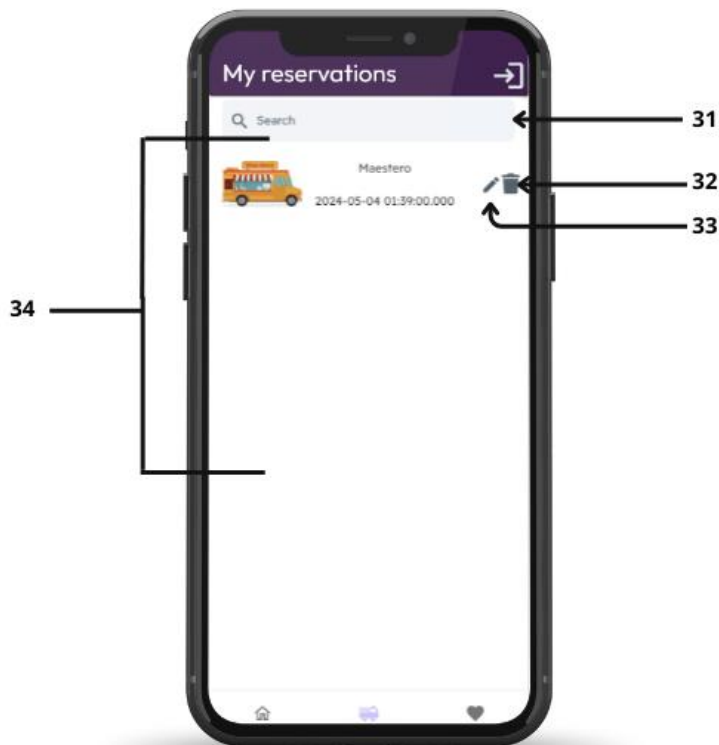


Figure 4-13: Edit/Delete/Search page.

#	Description
31	The user has the ability to search for reservation.
32	The user has the ability to delete a reservation by clicking on 'delete' button, then shows confirmation messages to make sure the user wants to delete reservation or not.
33	The user has the ability to edit a reservation by clicking on 'edit' button that drives the user to "Edit reservation" page.
34	Displayed the food trucks reservations with its name, date and time.

10. My Reservations page 'search' (if user click on #31):

Table 4-14: Search reservation

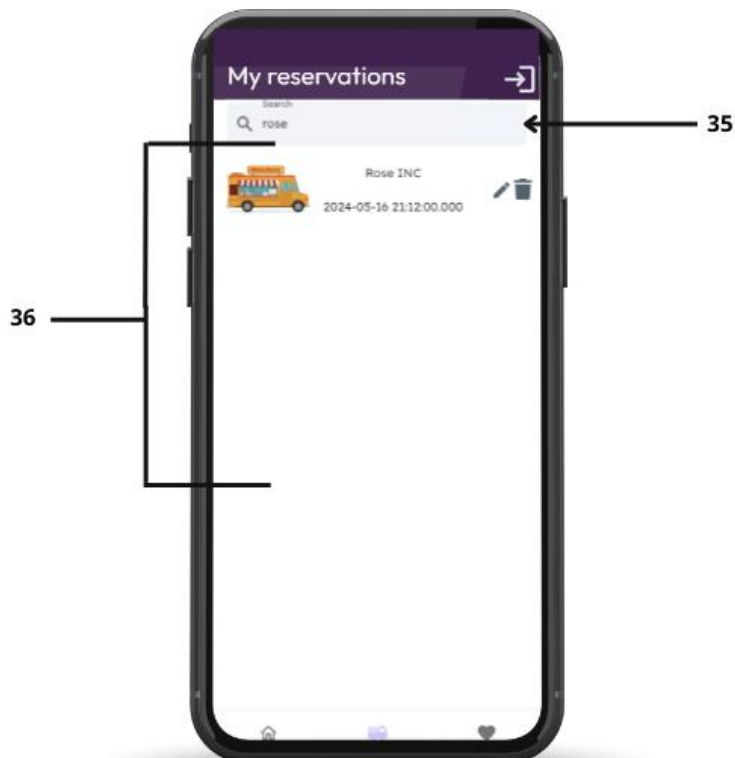


Figure 4-14: Search reservation.

#	Description
35	The user has the ability to search for reservation by writing the name of food truck.
36	Displayed the food truck reservation that has been searched with its name, date and time.

11. My Reservations page 'Delete Confirmation Message' (if user click on #32):

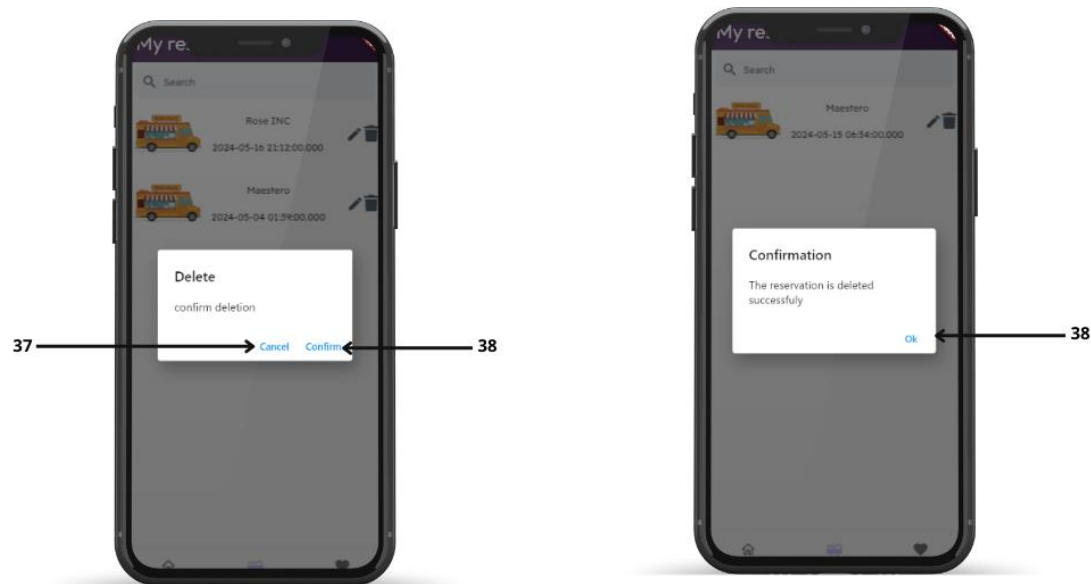


Figure 4-15: Delete Confirmation Message.

Table 4-15: Delete Confirmation Message

#	Description
37	Clicking on Cancel will close the confirmation message without deleting reservation.
38	Clicking on Confirm will pop-up another message to inform the user about success deletion, Clicking on Ok will close the message with deleting the reservation.

12. My Reservations page 'Delete (if user click on #38):

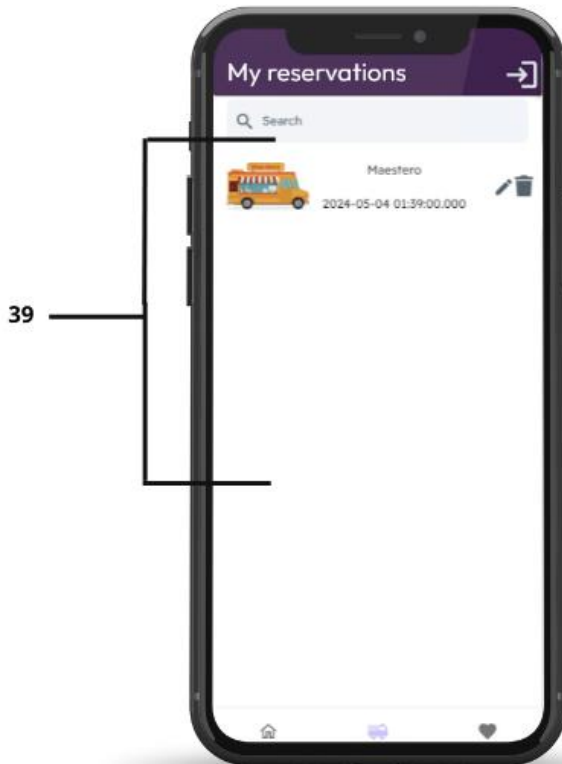


Figure 4-16: The reservation is deleted

Table 4-16 The reservation is deleted

#	Description
39	The deleted reservation will disappear from my reservations page.

13. Edit Reservation page (if user click on #33):

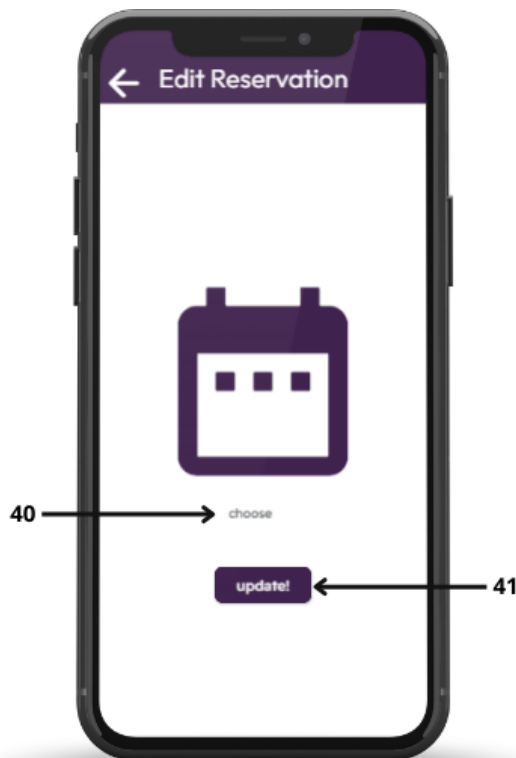


Figure 4-17: Edit reservation

Table 4-17: Edit reservation

#	Description
40	The user has the ability to click on 'choose' button that pop-ups a window for user to select date and time for his edited reservation.
41	The user has the ability to update a reservation by clicking on 'update!' button, then shows confirmation messages to make sure the user wants to update reservation or not.

14. Edit Reservation page 'Edit Confirmation Message' (if user click on #41):

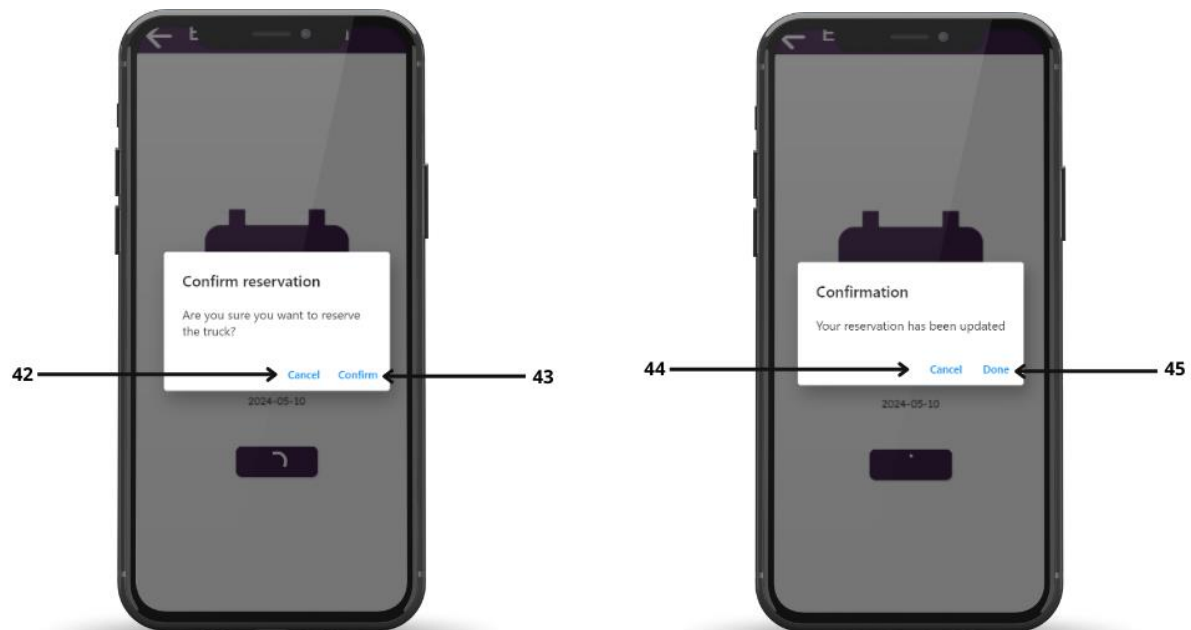


Figure 4-18: Edit Confirmation Message

Table 4-18: Edit confirmation Message

#	Description
42	Clicking on Cancel will close the pop-up window without editing.
43	Clicking on Confirm will confirm the reservation edit for the user and add the reservation changes to 'My Reservation' page and opening another confirm pop-up window.
44	Clicking on Cancel will close the pop-up window.
45	Clicking on Done will close the pop-up window with editing the reservation.

15. My Reservations page (if user click on #45):

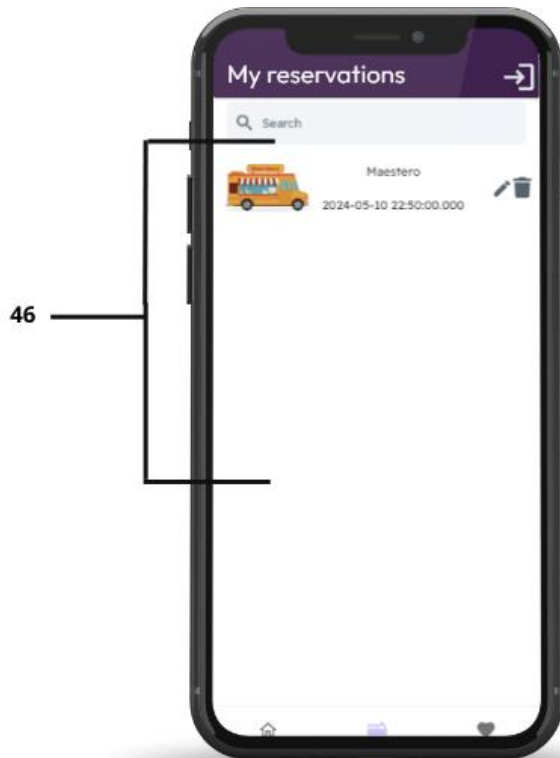


Figure 4-19: The reservation has been updated

Table 4-19: The reservation has been updated

#	Description
46	Displayed the food truck reservations with editing the date and time.

16. Wishlist page (if user click on #17):

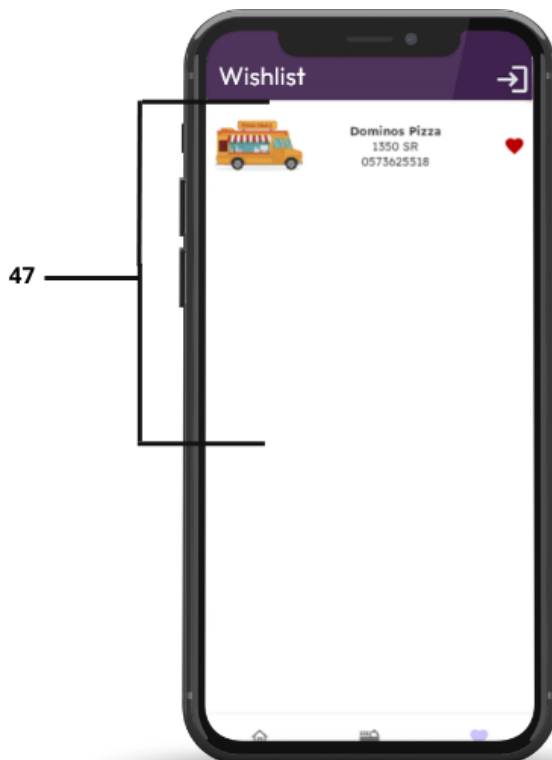


Figure 4-20: View foodtruck Wishlist

Table 4-20: View foodtruck Wishlist

#	Description
47	Displayed the static Wishlist food trucks with its name, price and phone number.

17. Home page (if user click on #13):

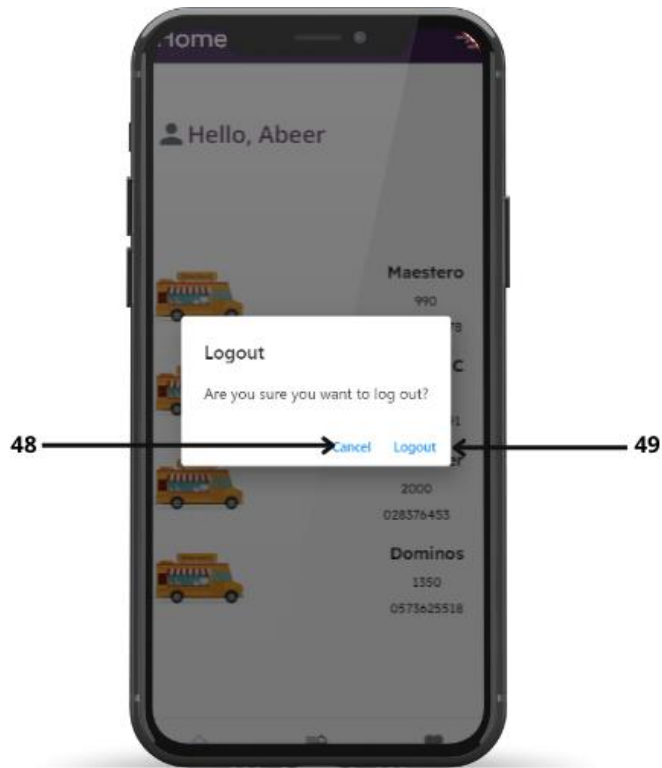


Figure 4-21: Confirmation Message for logout

Table 4-21: Confirmation Message for logout

#	Description
48	Clicking on Cancel will close the confirmation message without logging out.
49	Clicking on Logout will close the confirmation message with logging out to Main page.

- **Error Message:**
1. **Sign up without filling the required fields**

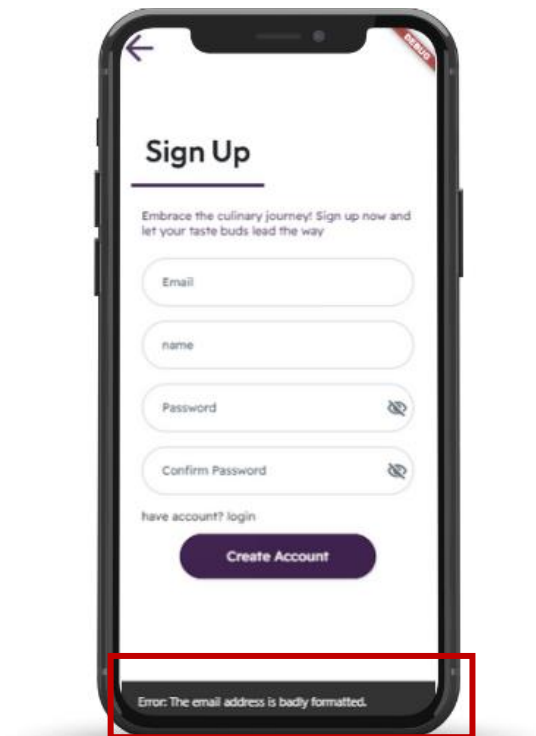


Figure 4-22: Error message to fill all the fields

2. **Sign up with credentials already used**

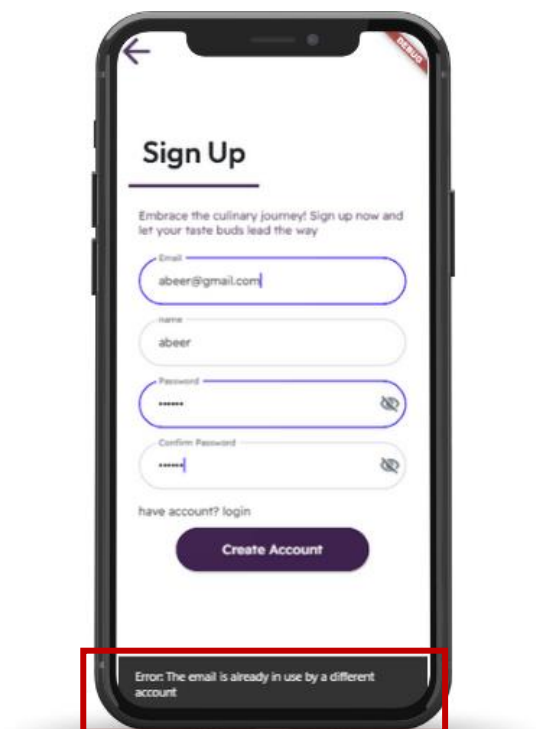


Figure 4-23: Error message to change the email

3. Sign up with password less than 6 digits

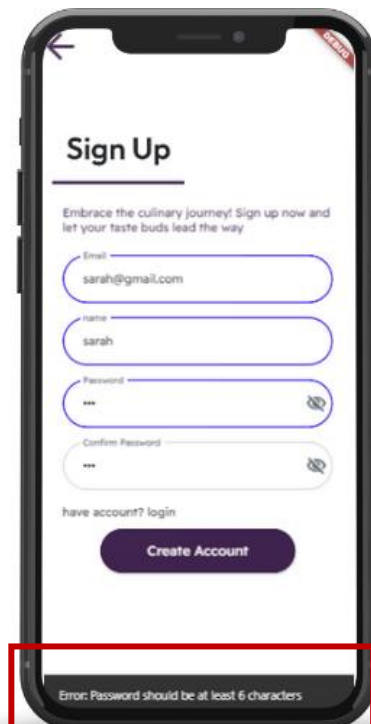


Figure 4-24: Error message to make the password 6 digits

4. Sign up with unmatched password

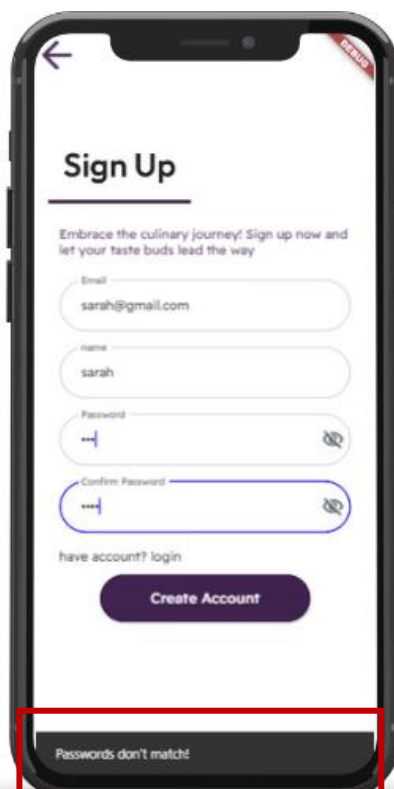


Figure 4-25: Error message to enter same passwords

5. Log in without filling the required fields

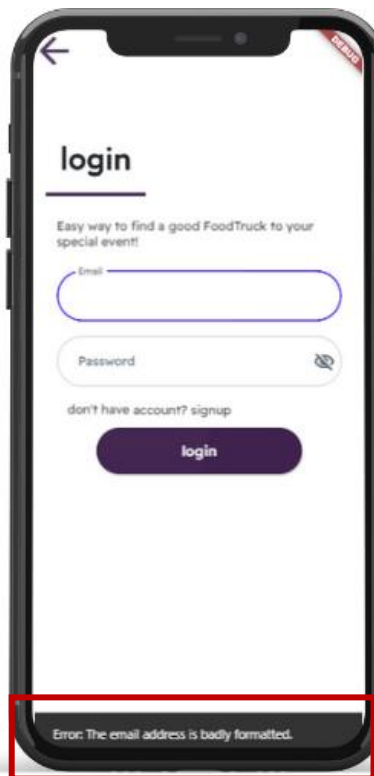


Figure 4-26: Error message to fill all the fields

6. Log in with wrong credentials

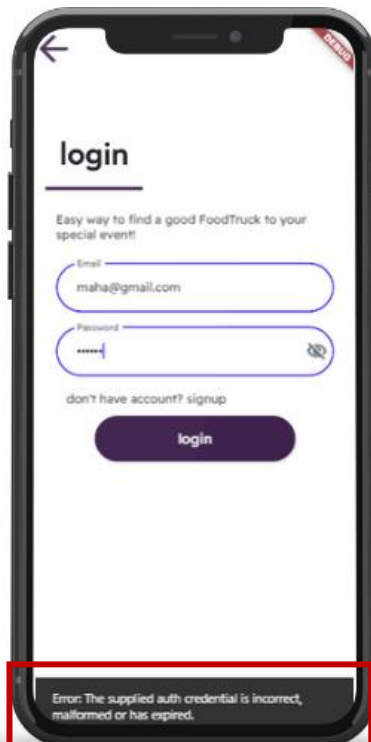


Figure 4-27: Error message to fill correct credentials

5 Chapter 5: Implementation

In this part, we will explain the implementation phase of our “Mutraf” app.

The implementation can be defined as creating and building the app, including the database and the functionalities of the app, in addition to linking the pages together and connecting the application flow. We started the project development in Android Studio; However, we have encountered many problems in this environment regarding passing the session variables between the pages. As a result, we decided to switch to FlutterFlow environment to optimize the development process. FlutterFlow has a seamless environment for both coding and creating interfaces.

The implementation included the following steps:

- 1- **Requirements gathering:** initially, we collected the necessary requirements to establish the basis for the development process. This involved comprehending the business goals, user needs, and technical specifications.
- 2- **Building User Interface:** we began the development by creating the user interfaces for the application, which involved creating sketches and wireframes for various screens and functionalities of the application.
- 3- **The main development process:** which involves building the application, including the front-end and back-end components:
 - Building the Front-end: we built the Front-End of the application using Dart programming language in FlutterFlow environment, this included:
 - a) Building user interfaces.
 - b) Creating all different screens in client-side that the user will interact with.
 - Developing the Back end: after building the front-end, we moved on to developing the Back-end of the application using Dart programming language and FireBase to build the database. The back-end development included:
 - a) Creating the database using FireBase website, then connect it to our project in FlutterFlow.
 - b) Connect the pages together -that were created in front-end-, to build the system flow.
 - c) Coding and implementing the functions of the app, i.e: Register, Log-in, Log-out, View all trucks in the system, Add reservation, Edit reservation, Delete reservation, View my reservations, Search my reservations logically and syntactically.
- 4- **Testing and debugging:** during the development process, we performed testing and debugging on both the front-end and back-end, including user acceptance testing, user story acceptance testing, and integration testing, to verify that the application’s functions work properly and that they fulfill user requirements.

As for the challenges, the main hurdles we have gone through in the code were the following:

- 1- The main challenge was concerning “edit” function, the process of sending the new information to update the reservation date and time was challenging in the code, but after few trials, we got the correct semantic of this function implementation. Also, almost all database interactions were to some extent challenging, to ensure proper configuration and synchronization of data and reservations across the application’s pages.

- 2- Validating user input. For example, rejecting invalid email format, and rejecting reserving a truck for a date that has passed already.

Our application consists of several key elements and components, such as the hardware and software components, including:

- 1- Front-end: The front-end of the application was built using Dart and it includes user interfaces and other elements (screen shapes) the users interact with.
- 2- Back-end: The back end of the application was built using Dart programming language and Firebase, it includes the implementation of all functions.
- 3- Database: The application used a non-relational database to store user data, trucks data, and reservation information. Our app is connected to the Firebase API to implement and connect to the database.

To facilitate the implementation process and ensure smooth collaboration, we utilized GitHub and Jira as hosting platforms. These platforms allowed us to upload our code and manage user stories.

In conclusion, the implementation phase of our “Mutraf” application included several steps and challenges to ensure proper and logical flow of operations. The result was a trusted and seamless Food-trucks reservation application that optimizes both the user experience with the application and the performance of the reservation process by addressing and delivering the users’ needs.

GitHub repository link: <https://github.com/Wissayifmk/MutrafApplication.git>

Jira project link:

<https://mutraf.atlassian.net/jira/software/projects/SCRUM/boards/1?atlOrigin=eyJpIjoiZjUzMzNkN2E1OTBhNDkxM2FiOWE4ZGM5NmZWNhZTUiLCJwIjoiaj9>

6 Chapter 6: Testing

In this chapter, we will conduct software testing, we will perform User story acceptance testing, Integration testing , and User acceptance testing, to ensure that our application meets both functional requirements and user expectations. These comprehensive testing methods are designed to guarantee a robust and user-friendly product.

6.1 User Story Acceptance Testing

We will begin with User Story Acceptance Testing, where each feature is evaluated against the agreed-upon user stories to ensure user needs fulfilment from the user prescriptive, by making sure that the acceptance criteria is met for each user story.

We first prepared the user stories to be tested, then conducted the testing with real users from our targeted users in prepared settings and environment. Then we observed the users while they performed the tests to ensure that the acceptance criteria were met.

All user stories were tested, which are related to: sign up, log in , log out, search reservations, wishlist , make a reservation , edit reservations , delete reservations, view trucks information, view reservations functions, in addition to non-functional requirements related user stories , such as speed and availability, detailed information about the user stories tested and there acceptance criteria will be provided in the following table.

Table 6-1: User Story Acceptance Testing

Sprint Number	User story	Acceptance criteria	Test action(s)	Pass ?	Comments
2	As a user, I want to be able to register and sign up so that I can use the application and its features.	- As a nonregistered user who wants to sign-up, if I go to signup page and enter my email, name, password that is more than 8 numbers, and click on sign up button, then I should be able to access the application. - As a nonregistered user if I go to the sign-up page and enter email that already exists and click on sign up button, then I should receive an error message. - As a non-registered user who wants to sign up, if I leave any required fields blank (e.g., email, password, name), the system should display an error message since all fields are mandatory and prompt me to fill in the missing information.	- Click on sign up option from the main application page. - Fill the signup fields, with invalid email, short password and a blank field. - Click on (Create account) button. - Verify that an error message is shown. - Fill all signup fields properly (Email, Name , Password, and confirm password) - Verify that the account is created, and that the user is directed to the Homepage (dashboard)	Yes	-

2	As a user, I want to log in with my username and password so that the system can authenticate me, and I can access my account and my information.	• If I go to the log-in page and enter my email and password and click on Log in, then the data associated to my user should be accessible. • If I go to the log in page and enter incorrect email or password and click on Log in, then log in fails with an error message that shows the email or password was wrong without specifying which is wrong for security. • If I go to the log in page and enter my email and password and click on Log in, then my user login session is loaded in less than eight seconds.	1- Click on log in option form the main application page 2- Fill the log in fields with invalid email, and incorrect password 3- Verify that an error message is shown(without specifying which field is incorrect) 4- Fill the log in fields with correct email and password 5- Verify that the user is authenticated and that the user is directed to the Homepage (dashboard) 6- Verify that the direction to the Homepage is done In less that 8 seconds	Yes	-

2	As a user, I want to be able to log out from my account so that I ensure the security of my account and personal information.	- As a logged-in user, if I click on the "Logout" button within the application, then I should be successfully logged out from my account. - If I logged out and try to access any protected pages or features within the application, then I should be redirected to the main page, indicating that I no longer have access until I log back in.	- Click on the logout button at top right corner of application screens - Verify a confirmation message is shown with two options ('logout', 'cancel') - Click the 'cancel' option in the pop-up message, verify that the pop-up message disappears, and the user remains in their account. - Click the log-out button again - Click the 'logout' option in the pop-up message, verify that the user is logged out of their account and no longer has access to the application. - Try to access a page within the application - Verify that the session is expired and the user must log back in.	Yes	-

2	As a user, I want to search for specific food trucks I reserved by its name so that I can easily find and access my preferred food truck.	- As a user, when I access the food truck search section in my reservations page in the app, I should be presented with a search bar or input field. - If I submit the search query, then the system should process the request and display the search results, focusing on food trucks that match the entered name. - If I submit the search, then the search results should prioritize exact matches or closely related matches to the entered name, ensuring that the preferred food truck is prominently displayed.	- Click on the search bar in my reservations page - Type the required truck name - Verify that the results appear in real time with exact matching trucks prioritized. - Verify that the desired truck is shown.	Yes	-

2	As a user, I want to make a reservation for a food truck or catering service of my choice so that I can secure dining arrangements and ensure availability	- As a user, if I access reservation section, then I should be presented with list of available food trucks or catering services. - If I have found a food truck or catering service, then I should be able to view detailed information about it, including pricing, and available dates/times for reservation. - As a user, if I select a specific date and time for my reservation, then the system should check the availability of the chosen food truck or catering service and display confirmation if it is available. - If the reservation is successfully made, then the system should provide me a confirmation message.	- Access dashboard page - Verify all available trucks in the system are shown in the dashboard. - Click on a desired food truck. - View all information about this specific food truck (name, price, phone Number, category, description, available time and date) - Select specific time and date for the reservation. - Verify that availability is checked. - Verify confirmation of available time and date (unavailable dates and times are gray and non-clickable) - Click "reserve" button - Verify that a confirmation message is displayed with two options "confirm and cancel"	Yes	-
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			10- Click “cancel” 11- Verify that reservation has not been made and the user is redirected to the dashboard 12- Repeat the above steps of making a reservation 13- Click on “confirm” rather than “cancel” in the confirmation message 14- Verify that a confirmation message that the reservation has been made and the user is redirected to dashboard		
2	As a user, I want to edit my existing reservation details (e.g., date, time) so that I can reflect necessary changes of my dining plans.	- As a user, if I go to reservation page, then the reservations should display with a button to edit the reservation details. - If I select the edit option for a reservation, then the system should display the current details of the reservation, including the date, time, food truck name, and any additional information. - If I go to the reservation page, then the system should allow me to modify the existing reservation details, such as the date, time.	1- Go to my reservations page 2- Verify that an “edit” button is available for each reservation 3- Click on “edit” button for a specific reservation 4- Verify that current time and date of the reservation are shown 5- Choose new date and time of the reservation	Yes	-

		• If the modified reservation details are valid and available, then the system should update the reservation with the new information • If the reservation is modified, then the system should provide a confirmation message indicating that the changes have been saved.	6- Verify that new time and date availability are checked (unavailable date and time are shown in gray and are not-clickable) 7- Click “Update” button 8- Verify that a confirmation message is shown with two option “confirm and cancel” 9- Click “cancel” 10- Verify that the reservation is not changed, and the user is redirected to reservations page 11- Repeat the above steps of editing a reservation 12- Click on “confirm” rather than “cancel” in the confirmation message 13- Verify that a confirmation message that the reservation has been edited and the user is redirected to reservations page		
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2	As a user, I want to delete my reservation if my plans change, so that I can free up the reserved slot and allow other users to make use of it.	• As a user, if I access my account, then I should be able to view a list of my existing reservations • If i go to the reservations in my account, them the system should displaye all reservations with a button to delete them. • If I select the delete option for a reservation, then the system should prompt for confirmation to ensure I intend to cancel the reservation. • If I confirm the deletion, then the system should remove the reservation from my account and free up the reserved slot for other users. • If i click on delete button, then the system should display a confirmation message indicating that the reservation has been successfully canceled.	1- Go to my reservations page 2- Verify that a “delete” button is available for each reservation 3- Click on “delete” button for a specific reservation 4- Verify that a confirmation message is shown with two options “confirm and cancel” 5- Click on “cancel” 6- Verify that the reservation is not deleted and the user is redirected to reservations page 7- Repeat the above steps of deleting a reservation. 8- Click on “confirm” option of the confirmation message rather that “cancel.” 9- Verify that the reservation is deleted, and the slot is available for other users	Yes	-
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			10- Verify that the user is redirected to reservations page.		
2	As a user, I want to view detailed information about food trucks, such as their cuisine and available hours, so that I can make informed decisions and choose the best option for my dining experience.	- As a user, if I access the food truck listing from the app, then I should be able to click on a specific food truck to view its detailed information. - If I go to the detailed information then the system should display relevant details about the food truck, including its name, cuisine category, contact information, and any additional information that helps users identify and understand the food truck.	1- Open the dashboard page in the app 2- Verify that all food trucks in the system are shown in the dashboard 3- Click on a specific food truck 4- Verify that all food-truck information is shown. i.e: (name, price, phone Number , category, description, available time and date)	Yes	-
2	As a user, I want to view my reservations within the application, including details such as the date, time, food truck name so that I can stay organized and keep track of my upcoming dining arrangements.	- As a user, if I access my account, then I should be able to see a list of my reservations. - If I go to the reservations page, then the system should display essential information for each reservation, including the date, time, and food truck name. - If I select a reservation from the list, then the system should provide a detailed view that includes all pertinent information about the reservation.	1- Open reservations page from the navigation bar 2- Verify that all my reservations are shown in the page 3- Verify that the reservation details (truck name, date and time) are shown 4- Navigate between reservations	Yes	-

		- If there are any updates or changes to a reservation, such as a modified date or time, then the system should reflect the updated information in the reservation view. - As a user, if I go to reservations page, then I should be able to navigate between different reservations and switch between the list view and detailed view effortlessly with a clear and organized manner.	5- Verify that the navigation between the reservations is done easily and effortlessly 6- Verify that the page reflect changes in reservations (such as deletion and edit)		
2	As a user, I want the system to be available 95% of the time when I try to access it, so that I don't get forced to search for other applications to use.	- As a user, if I opened the application 10 times a day, then 9 of them should be available to use.	1- Open the application 10 times. 2- Verify that at least 9 of 10 times, the application is available.	Yes	-
2	As a user, I want the search results for available food trucks to be displayed within 5 seconds after submitting desired option, ensuring a fast and responsive experience.	If I submit a search query for food trucks, then the system should process the request and display the search results within 5 seconds.	1- Open reservation page from the app 2- Click on the search bar 3- Search for a specific truck by name 4- Verify that the result are shown by at most 5 seconds	Yes	-

As shown in the table above, all the user-stories acceptance criteria were met, this indicates that the system reflects user needs and requirements and fulfils what is sought from it.

6.2 Integration Testing

In this section, we will use integration testing to examine the interactions that occur when components are combined into a system.

The next testing we will perform is the integration testing. Integration testing checks how individual components interact when integrated into a system. It is conducted each time a new component is added to the system, and it focuses on the interface between the system components, to verify that different components of the application work seamlessly together, ensuring a smooth, integrated user experience.

6.2.1 Integration Hierarchy

In this sub-section, we will present “Mutraf” integration hierarchy graph, which is designed to show the structure and the sequence in which various components of the system are integrated and tested to form “Mutraf” application.

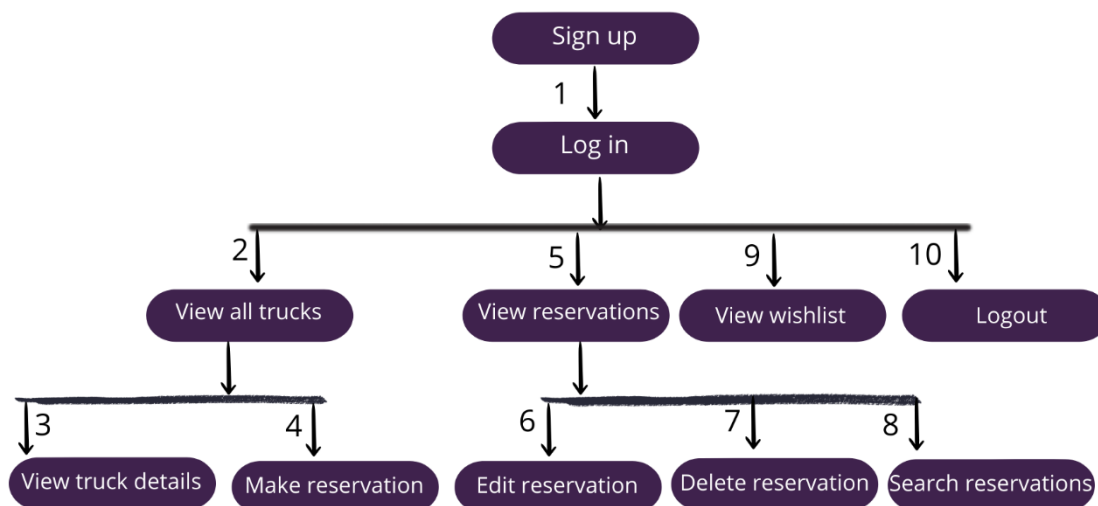


Figure 6-1: Integration hierarchy.

6.2.2 Integration Testing Plan

In this sub-section, we will present the integration testing plan table that is introduced to ensure that the system components interact with the new components properly by examining some test cases related to them, with an indication whether or not these components pass the test cases.

Table 6-2: Integration Testing

System components	New component	Test case	Pass ?	Comments
- Sign up to the system	Log in to the system	1. A user signed up into the system successfully. 2. The signed-up user logged in to the system	Yes	-
- Sign up to the system. - Log in to the system	View all trucks	1. A user signed up into the system successfully. 2. The signed-up user logged in to the system. 3. The logged-in user viewed all trucks in the application.	Yes	-
- Sign up to the system. - Log in to the system. - View all trucks	View truck details	1. A user signed up into the system successfully. 2. The signed-up user logged in to the system. 3. The logged-in user viewed all trucks in the application. 4. The logged-in user viewed a specific truck detail from the truck list.	Yes	-
- Sign up to the system. - Log in to the system. - View all trucks. - View truck details	Make a reservation	1. A user signed up into the system successfully. 2. The signed-up user logged in to the system. 3. The logged-in user viewed all trucks in the application.	Yes	-

		- The logged-in user viewed a specific truck detail from the truck list. - The logged-in user made a reservation for a specific truck for a specific date and time.		
- Sign up to the system. - Log in to the system. - View all trucks. - View truck details - Make a reservation	View reservations	- A user signed up into the system successfully. - The signed-up user logged in to the system. - The logged-in user viewed all trucks in the application. - The logged-in user viewed a specific truck detail from the truck list. - The logged-in user made a reservation for a specific truck for a specific date and time. - The logged-in user Viewed all their reservations in reservations page.	Yes	-
- Sign up to the system. - Log in to the system. - View all trucks. - View truck details - Make a reservation. - View reservations	Edit a reservation	- A user signed up into the system successfully. - The signed-up user logged in to the system. - The logged-in user viewed all trucks in the application from the homepage. - The logged-in user viewed a specific truck detail from the truck list.	Yes	-

		- The logged-in user made a reservation for a specific truck for a specific date and time. - The logged-in user Viewed all their reservations in reservations page. - The logged-in user edited a reservation he made previously, by modifying the date and time.		
- Sign up to the system. - Log in to the system. - View all trucks. - View truck details - Make a reservation. - View reservations - Edit a reservation	Delete a reservation	- A user signed up into the system successfully. - The signed-up user logged in to the system. - The logged-in user viewed all trucks in the application. - The logged-in user viewed a specific truck detail from the truck list. - The logged-in user made a reservation for a specific truck for a specific date and time. - The logged-in user Viewed all their reservations in reservations page. - The logged-in user edited a reservation he made previously, by modifying the date and time. - The logged-in user deleted a reservation he has made previously.	Yes	-

- Sign up to the system. - Log in to the system. - View all trucks. - View truck details - Make a reservation. - View reservations - Edit a reservation. - Delete a reservation	Search reservations	1. A user signed up into the system successfully. 2. The signed-up user logged in to the system. 3. The logged-in user viewed all trucks in the application. 4. The logged-in user viewed a specific truck detail from the truck list. 5. The logged-in user made a reservation for a specific truck for a specific date and time. 6. The logged-in user Viewed all their reservations in reservations page. 7. The logged-in user edited a reservation he made previously, by modifying the date and time. 8. The logged-in user deleted a reservation he has made previously. 9. The logged-in user searched his reservations by name, in reservations page.	Yes	-
- Sign up to the system. - Log in to the system. - View all trucks. - View truck details - Make a reservation. - View reservations - Edit a reservation. - Delete a reservation.	Log out from the system	1. A user signed up into the system successfully. 2. The signed-up user logged in to the system. 3. The logged-in user viewed all trucks in the application.	Yes	-

- Search reservations		4. The logged-in user viewed a specific truck detail from the truck list. 5. The logged-in user made a reservation for a specific truck for a specific date and time. 6. The logged-in user Viewed all their reservations in reservations page. 7. The logged-in user edited a reservation he made previously, by modifying the date and time. 8. The logged-in user deleted a reservation he has made previously. 9. The logged-in user searched his reservations by name, in reservations page. 10. The logged-in user logs out of the system and the session is ended.		
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As seen in the table 6-3, all the system components have passed the test cases successfully, indicating that the components in the system interact with each other seamlessly and that they achieve the functionality needed by the user.

6.3 User Acceptance Testing

The last testing to be performed is the user acceptance testing, which allows real users from the targeted audience to test the application in real-world scenarios, ensuring that it meets their requirements and performs effectively in real life situations, and it's also used to check if the system fulfills business requirements.

To carry out this test, we created a team of **eight** users, all of whom had basic proficiency in using mobile apps and fit the profile of our intended audience, specifically being aged 18 and older and interested in food trucks. The test was held in a quiet room.

They were given a scenario that covered all the functionalities of the app including signing up, logging in, viewing all food-trucks, view food truck details, reserve a food truck, view reservations, search reservations, edit reservations, delete reservations, and lastly, logging out. Prior to the test, we gave the users three minutes to familiarize themselves with the app. Following this, the test was performed based on the provided scenario. Their feedback was gathered via a questionnaire of five questions, where they evaluated their experience and pointed out any functional flaws or potential enhancements. Lastly, we analyzed the feedback to pinpoint usability problems.

6.3.1 Demographics of participants

In this sub-section, we will represent the participants' demographics, this includes the information we have gathered from the users regarding their age, income range and gender.

We will summarize the demographics results from the pie chart as follows:

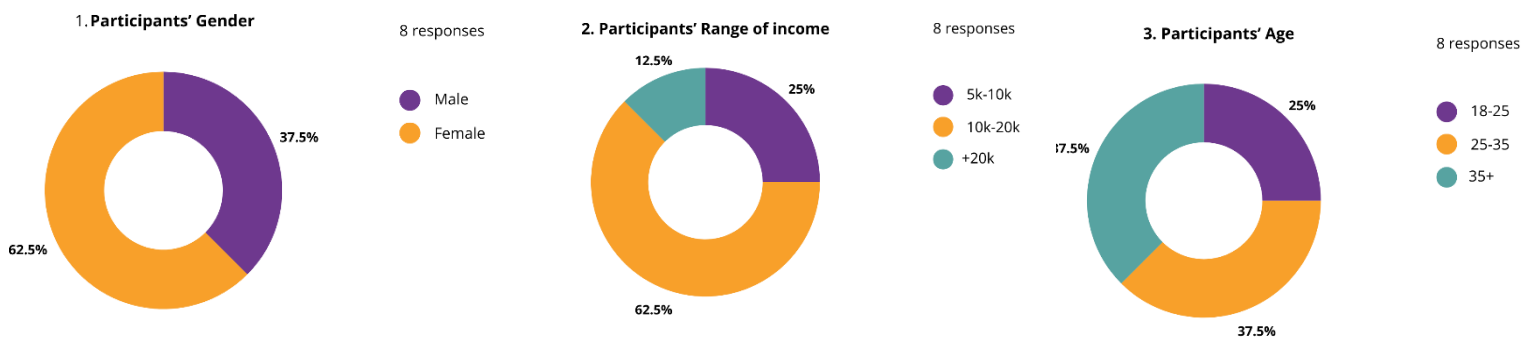


Figure 6-2: Demographics results

Gender Distribution:

The participant demographics show a gender distribution with approximately 62% female and 37% male. This indicates a relatively balanced representation of both genders. (See Graph 6.2)

Income Ranges:

About 25% of participants have an income ranging from 5,000 to 10,000 Riyals.

The majority, which approximately 62%, earn between 10,000 and 20,000 Riyals monthly.

And Around 12% of participants report a monthly income of over 20,000 Riyals. (See Graph 6.2)

Age Groups:

An equal percentage of participants are within the 25-35 year age group, and those older than 35 years. The minority however, which is about 25% of participants, are aged between 18 and 25 years. (See Graph 6.2)

These graphical representations provide a clear view of the demographics of the participants involved in the testing.

6.3.2 Questionnaire

In this sub-section, we will provide the participants' answers to the questions we asked them related to their feedback on the system and testing process based on the provided scenario. As we mention before, we asked eight participants who perform our tasks and give us some feedback about the system and testing process.

Questionnaire Questions:

- Was the process of signing up and logging in seamless?
- Were you able to complete the designated tasks without encountering any errors?
- Was it easy to navigate between the different pages of the app?
- Did the software meet your expectations in terms of performance during this scenario?
- Will you use the app again in the future?

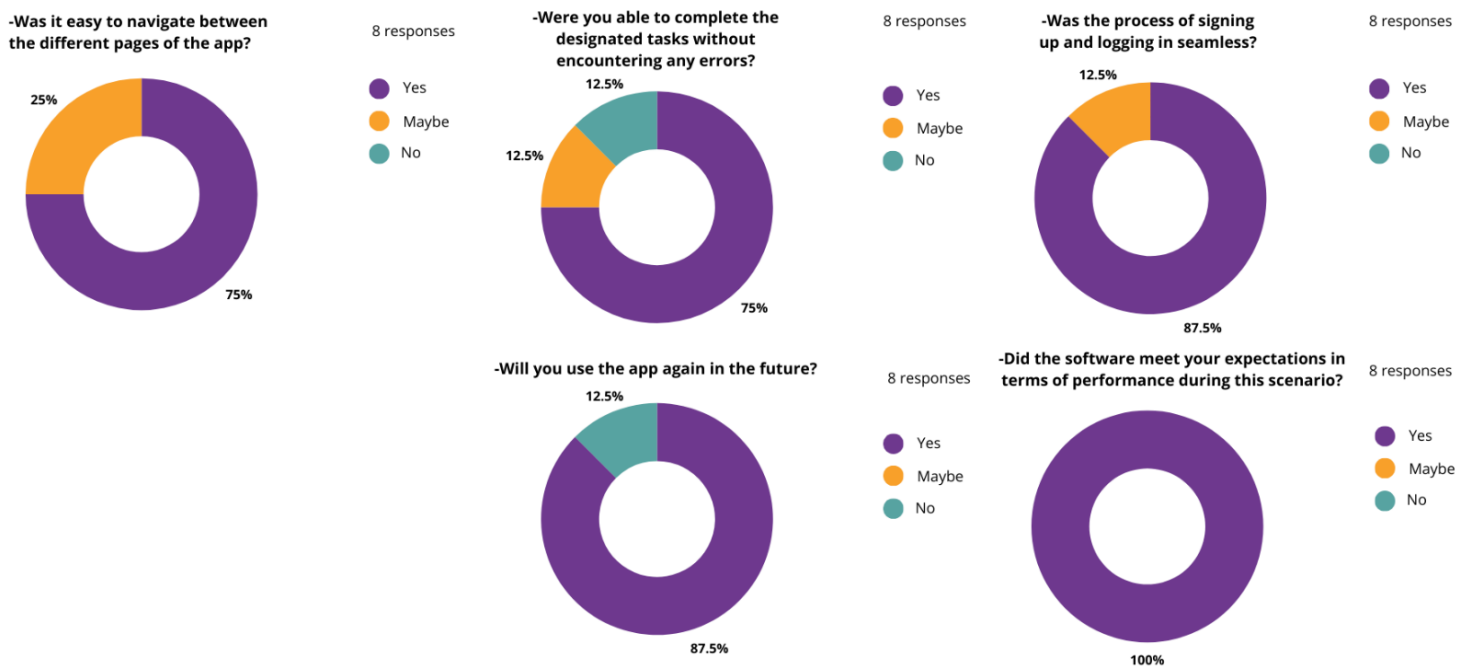


Figure 6-3: participants' answers to the questions related to their feedback on the system

We will look at the results obtained from the questionnaire:

Navigation Ease:

As shown in figure 6.3 the majority (75%) of the users found it easy to navigate between the different pages of the app. While 25% of the users expressed hesitation, improvement in the navigation interface will be considered.

Error-free scenario Completion:

The majority (75%) of the users reported their ability to complete the designated tasks without encountering any errors, indicating good functionality and performance (See Graph 6.3).

However, 12.5% encountered issues, and another 12.5% were uncertain, indicating potential improvement in the application's functionality.

Future Usage Intent:

Most users (87.5%) expressed a willingness to use the app again in the future, reflecting positive user satisfaction and likelihood of repeated usage. Only a small fraction (12.5%) didn't want to use the application in the future, which might be due to errors or performance issues they have encountered (See Graph 6.3).

Signup and Login Process:

A significant majority (87.5%) found the process of signing up and logging in to be seamless, indicating an effective user process. The remainder (12.5%) had a neutral response, suggesting that while not problematic, the process could be enhanced to increase user satisfaction (See Graph 6.3).

Performance Satisfaction:

As shown in figure 6.3, All respondents (100%) felt that the software met their expectations in terms of performance during the testing scenario. This is a strong indicator of the "Mutraf" reliability.

Overall, the feedback is positive, particularly in terms of performance, navigation, and error-free processes. As it's clear application and easy to use, so they will recommend it to friends and family.

7 Chapter 7: Conclusion and Future Work

7.1 Conclusion

In conclusion, our "Mutraf" document presents a comprehensive overview of the software product release. We address the problem statement, solution, and the product itself, including its vision, roadmap, objectives, and scope. We also highlight the Scrum team responsible for the development process. Additionally, we cover domain analysis, encompassing terminology, general domain knowledge, customers and users, the environment, tasks, and procedures, as well as competing software in the market.

The requirements engineering section identifies system users, presents the use case diagram, and defines the product backlog along with the criteria for the definition of done.

Furthermore, we explore system design aspects such as system architecture, class diagram, data design, component design, and interface design.

We also touch upon the implementation phase and provide insights into the testing process, including user story acceptance testing, integration testing, and user acceptance testing. The user acceptance testing section includes information on participant demographics and a questionnaire to gather valuable feedback.

Overall, our document serves as a comprehensive guide, offering a thorough understanding of the problem, solution, product, domain analysis, requirements, system design, implementation, and testing. It lays the foundation for a successful software product release and paves the way for future iterations and enhancements of "Mutraf".

Lastly, the output of this project is reliable software that addresses the needs of the users and serves as a third party between truck owners and customers, providing a seamless food-trucks reservation mobile application.

7.2 Future Work

In this section, we will suggest some of our plans related to improving “Mutraf” app in the future.

Firstly, tracking and notification system could be integrated to allow users to track food trucks in real-time and receive updates about their current locations and estimated arrival times, this feature would improve the user experience. Furthermore, a Wishlist feature might be added and enabled, to allow users to mark their preferred food trucks for later reservations.

Secondly, Integrating a secure payment system can optimize the payment process and support user convenience, so that users can easily make transactions for their reservations without the need for additional platforms or manual processes.

Thirdly, providing a customer review and rating system would provide valuable feedback for both users and food truck owners. Users can share their experiences, rate the service, and leave comments, which would help other users make informed decisions when selecting a food truck. Food truck owners can use this feedback to improve their trucks and notice any areas for improvement.

Finally, multilingual capability would interest many users by allowing them to use the application in their native language. This feature would improve usability and accessibility.

By considering these next areas of development, "Mutraf" will be able to adapt to the changing needs of its users, stay competitive, and keep up with the rapidly changing food reservation systems market.

8 References

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8.1 Appendix A: Interview

This Appendix will represent all the interview-related supplements, including the interview questions and transcriptions.

- Interview Questions:

1. What is your favorite way to serve food at your special events?
2. Tell me about the difficulties you encounter when looking for a good food truck for your event?
3. Describe the procedures when searching for a good food truck until you confirm reservation?
4. If you encounter circumstances that force you to cancel your reservation, how do you see the cancellation procedures?
5. If there is an application that gathers food trucks, what features would you like to include?

- Interview's Transcriptions:

The interview transcriptions hold significant value as they provide a detailed record of the conversations between the interviewees and interviewers. They include essential information such as the names of the participants, interview location, time and date, objectives of the interview, agenda, observations made during the interview, and any topics that were not covered. Each transcription also presents five thought-provoking questions along with the interviewees' answers, allowing for in-depth insights and analysis. These transcriptions serve as a comprehensive resource for understanding the perspectives, experiences, and strategies discussed during the interviews.

Online Interview (1)	
Interviewee: Abdullah	Interviewer: Albandy
Location/Medium: Phone Call Phone Number: 0548363968	Appointment Date: 29Feb 2024 Start Time: 5:00PM End Time: 5:38:25PM
Objectives: - Gain an idea of people's preferences when dealing with food trucks. - Collect information about the challenges people may face when reserving food trucks.	Reminders: the interviewee had previously rented a food truck at his home, and had problem with it.
Agenda: - Introduction - Background in project - Overview of interview - Topic to be covered - Permission to record - Question1 - Question2 - Question3 - Question4 - Question5 - Summary of major points - Questions from interviewee - Closing	Approximate Time: 3min 4min 2min 1min 25sec 3min 5min 7min 3min 4min 1min 3min 2min
General Observations: He appeared to have a good understanding of the concept, as evidenced by his enthusiasm for our program, reflected in his body language and the change in his voice throughout the course of the interview.	
Topic not covered: Unfortunately, due to time limitations, we did not have the opportunity to delve into the specifics of the app with the interviewee.	

Figure 8-1 :Interview's Transcription for first Interviewee.

Online Interview (1)	
Interviewee: Abdullah	Date: 29Feb 2024
Questions:	Answers and Notes:
Q1: What is your favorite way to serve food at your special events?	I prefer having food stations at my special events where guests can interact directly with the food truck chefs and have their food prepared fresh on the spot. Observations: The interviewee expressed his preference for food stations at special events. his tone suggested that this is a method they truly enjoy and find appealing.
Q2: Tell me about the difficulties you encounter when looking for a good food truck for your event?	One difficulty I face when looking for a good food truck for my event is finding one that can accommodate dietary restrictions or special requests from my guests. Observations: When discussing the difficulties encountered in finding a suitable food truck, the interviewee seemed to shed light on some factors that the interviewer may not have previously considered. This indicated a unique perspective and provided new insights.
Q3: Describe the procedures when searching for a good food truck until you confirm reservation?	When searching for a good food truck, I typically rely on recommendations from friends or event planners. I then contact the food trucks to discuss availability, pricing, and menu options. Once I'm satisfied, I confirm the reservation by providing a deposit. Observations: The interviewee appeared to take his time in considering the features he would like to include in a food truck application. This suggested that he has specific needs and preferences, particularly in terms of an easier rental process and a focus on convenience.
Q4: If you encounter circumstances that force you to cancel your reservation, how do you see the cancellation procedures?	If I need to cancel my reservation, I believe there should be a clear cancellation policy in place, preferably with a reasonable refund or rescheduling option. Observations: The interviewee expressed a preference for a cancellation policy that offers a reasonable refund or rescheduling option. It appeared that he value flexibility and fairness in dealing with unforeseen circumstances that may require cancellation.
Q5: If there is an application that gathers food trucks, what features would you like to include?	If there is an app that brings food trucks together, I would like to see features like collecting only trucks that are in Riyadh, a sorting system for trucks based on prices, and the ability to pre-reserve a truck. for faster service. Observations: The interviewee's suggestion of having an app that collects only food trucks in Riyadh indicates a specific location preference. This observation suggests that the interviewee is focused on finding food trucks that are available within his desired area.

Figure 8-2 :Another Interview's Transcription for first Interviewee.

Online Interview (2)	
Interviewee: Maha	Interviewer: Albandry
Location/Medium: Maha's house, Riyadh, Saudi Arabia	Appointment Date: 28Feb 2024 Start Time: 6:00PM End Time: 6:40:30PM
Objectives: - Gain an idea of people's preferences when dealing with food trucks. - Collect information about the challenges people may face when reserving food trucks.	Reminders: the interviewee had previously rented a food truck at her home.
Agenda: Introduction Background in project Overview of interview Topic to be covered Permission to record Question1 Question2 Question3 Question4 Question5 Summary of major points Questions from interviewee Closing	Approximate Time: 3min 4min 2min 2min 30sec 5min 3min 6min 5min 4min 1min 3min 2min
General Observations: During the interview, the interviewee was excited and interested in answering the questions and sharing her thoughts. She also showed full support for the idea of creating this application.	
Topic not covered: Due to time constraints, we were unable to discuss the details of the app with the interviewee.	

Figure 8-3: Interview's Transcription for second Interviewee.

Online Interview (2)	
Interviewee: Maha	Date: 28Feb 2024
Questions:	Answers and Notes:
Q1: What is your favorite way to serve food at your special events?	My favorite way to serve food at special events is through a buffet-style setup where guests can choose from a variety of dishes and customize their own plates. Observations: When discussing their favorite way to serve food at special events, the interviewee seemed to have a preference for a buffet-style setup. She appeared confident and knowledgeable about this approach.
Q2: Tell me about the difficulties you encounter when looking for a good food truck for your event?	One difficulty I encounter when looking for a good food truck for my event is finding one that aligns with the desired cuisine and theme, as well as ensuring they have availability on the desired date. Observations: The interviewee highlighted the difficulties she face when searching for a food truck for her event. Her tone suggested that it can be a challenging task, perhaps indicating past experiences that didn't meet her expectations.
Q3: Describe the procedures when searching for a good food truck until you confirm reservation?	When searching for a good food truck, I usually start by researching online, checking reviews, and reaching out to different vendors for pricing and menu options. Once I find a suitable option, I confirm the reservation by signing a contract and paying a deposit. Observations: While describing the procedures for searching and confirming a reservation with a food truck, the interviewee provided detailed steps and seemed very organized.
Q4: If you encounter circumstances that force you to cancel your reservation, how do you see the cancellation procedures?	If I have to cancel my reservation due to unforeseen circumstances, I expect clear communication from the food truck and a fair refund policy. It's important to have a cancellation procedure that is straightforward and transparent. Observations: When discussing cancellation procedures, the interviewee appeared to have thought about potential circumstances that might lead to cancellation. she emphasized the importance of clear communication and a fair cancellation policy, indicating a desire for a hassle-free process.
Q5: If there is an application that gathers food trucks, what features would you like to include?	If there was an application that gathers food trucks, I would like it to include features such as Wishlist of food trucks, menu descriptions and photos, and the ability to make reservations and pre-reserve food trucks. Observations: The interviewee's response to the features they would like to see in a food truck application was thoughtful and well-considered. It seemed like she had given significant thought to the features that would enhance her experience, potentially based on previous challenges or frustrations she encountered.

Figure 8-4: Another Interview's Transcription for second Interviewee.

8.2 Appendix B: Questionnaires

This Appendix will represent all the questionnaire-related supplements, including the questionnaire questions, and a representation of the responses we obtained through pie charts. [7]

• Questionnaires Questions:

1. Do you think finding a good truck for your needs - ex: type of food, budget, etc....- is easy nowadays?

-Yes
-No
-Maybe

2. Do you prefer services that deliver food at your place?

-Yes
-No
-Maybe

3. Would you like to use an application that offers all food trucks in one place?

-Yes
-No
-Maybe

4. Would you like the option of pre-reserving the food trucks to save time and avoid waiting in long lines?

-Yes
-No
-Maybe

5. Do you enjoy the experience of trying different cuisines from a variety of food trucks in Riyadh?

-Yes
-No
-Maybe

6. Would you like the option of sorting the food trucks based on prices?

- Yes
- No
- Maybe

7. Would you like the feature of "Wishlist" for food trucks to save searching time, and provide more organization?

- Yes
- No
- Maybe

• Questionnaires Answers:

1. Do you think finding a good truck for your needs - ex: type of food, budget, etc....- is easy nowadays?

22 responses

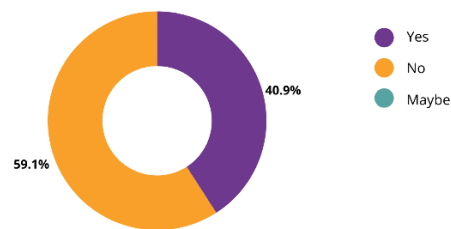


Figure 8-5: The pie chart of first questionnaire question.

2. Do you prefer services that deliver food at your place?

22 responses

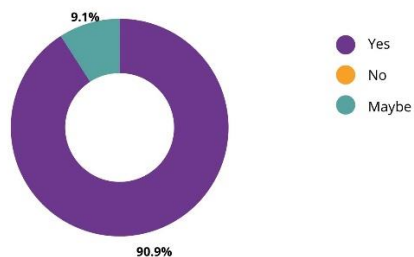


Figure 8-6: The pie chart of second questionnaire question.

3. Would you like to use an application that offers all food trucks in one place?

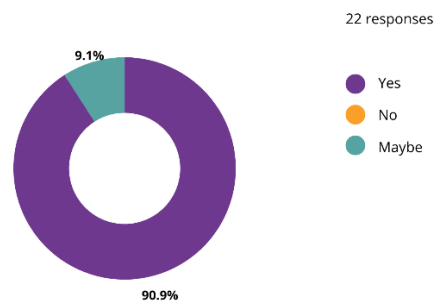


Figure 8-7: The pie chart of third questionnaire question.

4. Would you like the option of pre-reserving the food trucks to save time and avoid waiting in long lines?

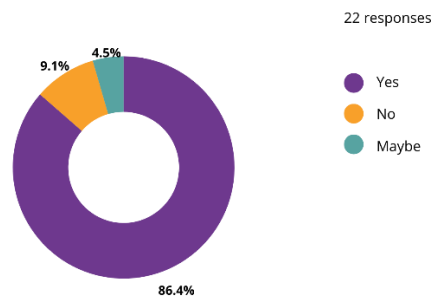


Figure 8-8: The pie chart of fourth questionnaire question.

5. Do you enjoy the experience of trying different cuisines from a variety of food trucks in Riyadh?

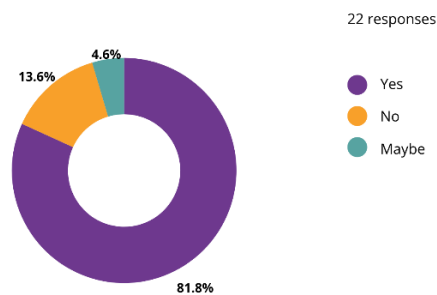


Figure 8-9: The pie chart of fifth questionnaire question.

6. Would you like the option of sorting the food trucks based on prices?

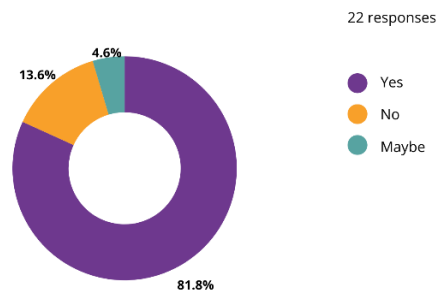


Figure 8-10: The pie chart of sixth questionnaire question.

7. Would you like the feature of "Wishlist" for food trucks to save searching time, and more organizing?

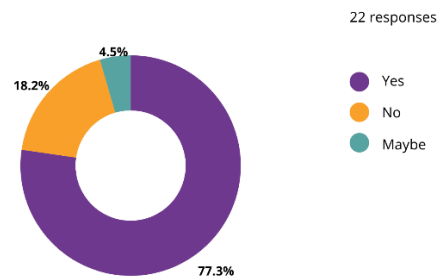


Figure 12: The pie chart of seventh questionnaire question.

8.3 Appendix C: Jira

This Appendix will represent the use of Jira, a popular project management tool, that we have utilized in our project workflow. Jira provides a comprehensive platform for managing tasks, tracking progress, and collaborating with team members. We will focus on two key aspects of Jira: The Product Backlog and Sprint 2. The Product Backlog serves as a dynamic list of prioritized tasks we intend to complete, while Sprint 2 represents a specific time-bound iteration within our project. [8]

• Product Backlog

Projects / My Scrum Project

Backlog

Search: [WA] [User Icons] Epic ▾

Import work Insights View settings

▼ Backlog (13 issues) 29 0 0 Create sprint

Issue Key	Description	Priority	Value	Assignee
SCRUM-1	As a user, I want to be able to register and sign up so that I can use the application and its features.	TO DO ▾	3	[User Icon]
SCRUM-2	As a user, I want to log in with my username and password so that the system can authenticate me and I can acces...	TO DO ▾	2	[User Icon]
SCRUM-3	As a user, I want to be able to log out from my account so that I ensure the security of my account and personal inf...	TO DO ▾	2	[User Icon]
SCRUM-7	As a user, I want to search for specific food trucks by name so that I can easily find and access my preferred food t...	TO DO ▾	2	[User Icon]
SCRUM-8	As a user, I want to sort food trucks based on various criteria (e.g. newest, popularity, cuisine type) so that I can fin...	TO DO ▾	2	[User Icon]
SCRUM-9	As a user, I want to create a wishlist of food trucks for easy access and future reference, so that I can conveniently ...	TO DO ▾	1	[User Icon]
SCRUM-4	As a user, I want to make a reservation for a food truck or catering service of my choice so that I can secure dining ...	TO DO ▾	3	[User Icon]
SCRUM-5	As a user, I want to edit my existing reservation details (e.g., date, time) so that I can make necessary changes to ...	TO DO ▾	2	[User Icon]
SCRUM-6	As a user, I want to delete my reservation if my plans change, so that I can free up the reserved slot and allow othe...	TO DO ▾	2	[User Icon]
SCRUM-10	As a user, I want to view detailed information about food trucks, such as their cuisine and available hours, so that I...	TO DO ▾	1	[User Icon]
SCRUM-11	As a user, I want to view my reservations within the application, including details such as the date, time, food truck...	TO DO ▾	1	[User Icon]
SCRUM-12	As a user, I want the system to be available 95% of the time I try to access it, so that I don't get forced to search f...	TO DO ▾	5	[User Icon]
SCRUM-13	As a user, I expect the search results for available food trucks to be displayed within 5 seconds after submitting d...	TO DO ▾	3	[User Icon] ...

Figure 8-11 :Product Backlog of Mutraf Application in Jira

• Sprint 2

Backlog

W/A SA MA

Epic

Import work Insights View settings

▼ SCRUM Sprint 2 28 Mar – 18 Apr (13 issues) 29 0 0 Start sprint

SCRUM-1 As a user, I want to be able to register and sign up so that I can use the application and its features.	TO DO	3	SA
SCRUM-2 As a user, I want to log in with my username and password so that the system can authenticate me and I can access...	TO DO	2	W/A
SCRUM-3 As a user, I want to be able to log out from my account so that I ensure the security of my account and personal info...	TO DO	2	MA
SCRUM-7 As a user, I want to search for specific food trucks by name so that I can easily find and access my preferred food tr...	TO DO	2	W/A
SCRUM-8 As a user, I want to sort food trucks based on various criteria (e.g. newest, popularity, cuisine type) so that I can fin...	TO DO	2	MA
SCRUM-9 As a user, I want to create a wishlist of food trucks for easy access and future reference, so that I can conveniently ...	TO DO	1	W/A
SCRUM-4 As a user, I want to make a reservation for a food truck or catering service of my choice so that I can secure dining a...	TO DO	3	W/A
SCRUM-5 As a user, I want to edit my existing reservation details (e.g., date, time) so that I can make necessary changes to m...	TO DO	2	MA
SCRUM-6 As a user, I want to delete my reservation if my plans change, so that I can free up the reserved slot and allow other ...	TO DO	2	W/A
SCRUM-10 As a user, I want to view detailed information about food trucks, such as their cuisine and available hours, so that I ...	TO DO	1	W/A
SCRUM-11 As a user, I want to view my reservations within the application, including details such as the date, time, food truck ...	TO DO	1	SA
SCRUM-12 As a user, I want the system to be available 95% of the time I try to access it, so that I don't get forced to search fo...	TO DO	5	W/A
SCRUM-13 As a user, I expect the search results for available food trucks to be displayed within 5 seconds after submitting de...	TO DO	3	SA

Figure 8-12: Sprint 2 Backlog of Mutraf Application in Jira